Inflation and Labor Markets: A Bottom-Up View

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Abstract

U.S. inflation surged in 2021-22 and subsequently declined, largely as goods inflation receded, while services inflation remained elevated. This paper focuses on services inflation, excluding housing, to understand how tight labor markets shape price dynamics. In doing so, it fills a gap in a literature that has typically related labor market tightness either to wages or directly to aggregate inflation, but has not traced the full transmission mechanism. We combine proprietary microdata on service-sector wages with variation in tightness across Metropolitan Statistical Areas in a local-projection instrumental-variable model. We find a strong transmission from local labor market tightness to service-sector wage growth and, in turn, from wage growth to services inflation. This pass-through is attenuated in standard measures of core inflation and in headline inflation. The wage-price pass-through is state dependent, becoming stronger when labor markets are unusually tight.

JEL-Codes: E24, E31
Keywords: Inflation, Wages, Labor Market Conditions

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The views expressed in this paper are those of the authors and do not necessarily represent the views of, nor should they be attributed to, the IMF, its Executive Board, or IMF management. The authors are grateful to Philip Barrett, Nigel Chalk, Era Dabla-Norris, Davide Furceri, Gita Gopinath, Pierre-Olivier Gourinchas, Daniel Leigh, Bo Li, Sole Martinez Peria, Papa N'Diaye, and Rodrigo Valdes for valuable discussions and insightful comments and suggestions.

1 Introduction

U.S. inflation rose sharply after the COVID-19 pandemic (Figure 1), with divergent patterns between price trends in goods and services (Figure 2). Goods inflation surged first, as supply constraints coincided with a shift in demand from services to goods, before receding. Services inflation rose less sharply than goods inflation and proved more persistent. These dynamics have revived a long-standing debate on the role of labor market conditions in generating and sustaining inflation. Services are of particular relevance because they are labor intensive and predominantly non-tradable, making their prices more closely connected to local labor costs. Understanding whether and how tight labor markets translate into services inflation is therefore central to assessing the persistence of inflation and the role of labor market conditions in inflation dynamics. We find that local labor market tightness has a strong and state-dependent effect on services inflation, transmitted through wages: tightness raises service-sector wage growth, which in turn passes through to services prices, and this pass-through is substantially stronger when labor markets are especially tight, a pattern concentrated in services rather than in broader price aggregates.

A growing literature has examined whether the unusually tight U.S. labor market contributed to the post-pandemic inflation surge. Recent studies find evidence that labor market tightness, measured by the vacancy-to-unemployment ratio (V/U) or related indicators, is associated with higher inflation, although estimates differ substantially. For example, Ball, Leigh and Mishra (2022) attribute a significant portion of the inflation to labor market tightness,[1] while Bernanke and Blanchard (2025) argue that, contrary to initial concerns that inflation would be driven by overly tight labor markets, the inflation surge was largely explained by price shocks with a modest contribution from the labor market. Furthermore, Ball, Leigh and Mishra (2022, 2025) and Benigno and Eggertsson (2023) document evidence of a nonlinear relationship between labor market tightness and inflation, while Beaudry, Hou and Portier (2026) argue that the estimated nonlinearity could be sensitive to the inclusion of inflation expectations in the analysis.

Understanding how labor market tightness is linked to inflation requires tracing the transmission in its entirety. Some studies estimate a wage Phillips curve, asking how labor market tightness affects wage growth. Others estimate a price Phillips curve, asking how labor market tightness affects inflation directly. These approaches provide important pieces of the transmission mechanism, but neither traces the full transmission using plausibly exogenous local variation, as we do. A strong wage response to labor market tightness need not translate into a comparable increase in prices if firms absorb higher labor costs through lower margins, productivity gains, or other adjustments. Conversely, a reduced-form relationship between labor market tightness and inflation does not establish that wages are the channel through which labor market conditions affect prices. Bernanke and Blanchard (2025) explicitly emphasize the importance of the wage-price link to understand inflation dynamics and estimate it using aggregate data. We complement this approach by estimating the wage-price transmission using cross-sectional variation in local labor markets and focusing on the services sector. Our conceptual framework is closely related to the regional Phillips curve

[1] Ball, Leigh and Mishra (2025) extend the analysis to the most recent period and show that the labor market remains the largest estimated contributor to inflation above target.

literature.[2] By exploiting cross-sectional variation across local labor markets, we examine how demand-driven wage pressures translate into local services inflation while controlling for common aggregate shocks. This cross-sectional perspective is particularly useful for identifying the wage-price transmission, because it separates local labor-market-induced wage movements from aggregate movements in inflation expectations and other common shocks.

Focusing on services also helps address an important limitation of aggregate inflation measures: headline and core inflation combine components with very different sensitivities to local labor costs. Whereas local labor costs likely play a direct role in service prices as they are predominantly non-tradable, non-services prices often reflect goods and commodity price movements that are influenced by national or global conditions. A wage-price relationship that is economically important can therefore be difficult to detect in standard measures of core or aggregate inflation. This distinction is particularly relevant for the post-pandemic period, when large shocks to goods and commodity prices moved headline inflation independently of local labor market conditions.

Against this background, this paper develops a unified framework to trace the transmission from local labor market tightness to services inflation through wages. We ask three related questions. First, how strongly does local labor market tightness affect wage growth in the services sector? Second, to what extent does wage growth in the services sector pass through to services prices? Third, does this pass-through become stronger when labor markets are unusually tight? By answering these questions jointly, we distinguish the wage response to labor market conditions from the subsequent transmission of wage pressures to prices, allowing us to identify where persistence and nonlinearity arise in the labor market-inflation transmission mechanism.

Answering these questions requires both granular wage data and a source of plausibly exogenous variation in local labor market conditions. A key input into our analysis is a measure of service-sector wages from a proprietary data source, Homebase. The data allow us to measure local service-sector wages at the monthly and metropolitan statistical area (MSA) level, at the same frequency and granularity as labor market tightness and services inflation. Existing administrative and survey data generally do not provide this combination of frequency, geographic coverage, and sectoral detail.

We implement a local-projection instrumental-variable (LP-IV) approach to identify the transmission mechanism. In the first stage, we instrument service-sector wage growth using national industry-level demand shocks interacted with local industry exposure, following the intuition behind the tradable-demand instrument developed by Hazell et al. (2022). A distinctive feature of our instrument is that the national shocks are constructed from tradable industries only. Because the shocks originate in tradable industries, they are less likely to reflect unobserved local shocks to service-sector wages or prices, while the local exposure component allows them to generate differential labor market pressure across MSAs.[3] In the second stage, we use the wage variation induced by these local labor market shocks to estimate the pass-through from service-sector wage growth to services inflation. Because our instrument enters through wages specifically, this approach isolates the component of

[2]See Hazell et al. (2022) for a review of the literature and theoretical foundations of the regional Phillips curve.

[3]We use non-tradable and service sectors interchangeably in the context of our IV estimation as nearly all services in the Consumer Price Index (CPI) calculation are non-tradable (Johnson, 2017).

the price response that operates through the wage channel, rather than the overall effect of labor market conditions on prices, which may also include other transmission channels such as markups or hiring costs (Ball et al., 2022). The combination of cross-sectional variation, high-frequency wage data, and an IV strategy allows us to isolate local wage pressures while absorbing common aggregate conditions, an empirical setting that is difficult to obtain from conventional data.

Our analysis yields three main findings. First, labor market tightness has a large and statistically significant effect on service-sector wage growth. A one-unit increase in the constructed shift-share tightness index is associated with approximately 12.5 percentage points higher year-on-year wage growth over a 10-month horizon.

Second, wage growth has a substantial and persistent effect on services inflation. A one-percentage-point increase in service-sector wage growth raises services inflation by roughly 0.5 percentage points over the same horizon. Combining the two stages, a one-unit increase in the constructed shift-share index is associated with an increase of approximately 6 percentage points in year-on-year services inflation at the 10-month horizon through the wage channel, accounting for more than one-third of services inflation excluding housing across MSAs between Q3 2021 and Q1 2023.

Third, this transmission mechanism is state dependent, becoming stronger when labor markets are tight. While the first-stage relationship between labor market tightness and wage growth is broadly similar across labor market conditions, wage-price pass-through is substantially larger when V/U exceeds one, particularly over the 6- to 12-month horizon. Importantly, this nonlinearity is not evident when we replace services inflation with headline or core inflation, indicating that the nonlinear transmission from labor market tightness to prices is concentrated in the services sector.

These findings contribute to and are distinguished from the existing literature in several key respects. First, they identify the wage-price transmission mechanism underlying the relationship between labor market conditions and inflation. While recent work has separately documented the responses of wages and prices to labor market tightness, our estimates directly connect the two by isolating wage growth generated by local labor demand shocks and tracing its subsequent effect on services prices. Second, the results highlight the importance of inflation composition for assessing the role of labor markets in inflation dynamics. The strong wage-price pass-through we find for services is substantially attenuated for broader inflation measures at short and medium horizons, suggesting that the role of labor markets can be obscured when sectors with limited exposure to local labor costs are included in the inflation measure studied. Third, our evidence adds a new dimension to the debate over Phillips curve nonlinearity. The stronger pass-through in tight labor markets arises primarily through the wage-price margin and is concentrated in services, helping reconcile our findings with recent evidence that nonlinearities are weaker or absent in broader inflation measures. More broadly, our results suggest that understanding the inflationary consequences of tight labor markets requires tracing the transmission mechanism across sectors and across the distinct margins connecting labor market conditions, wages, and prices.

Our findings have important implications for monetary policy. In the years preceding the pandemic, services inflation was the dominant contributor to headline inflation in the United States, while goods prices rose only modestly or fell outright (Figure 2). The labor market tightened significantly after the pandemic, and while wage growth has since moderated, it

remains high relative to pre-pandemic levels. If high wage growth or tight labor market persists, our estimated pass-through implies continued upward pressure on services prices with a lag; our results suggest that the pass-through takes almost a full year to peak. Such dynamics may complicate the inflation outlook, especially in an environment prone to supply-side shocks due to geopolitical tensions, higher tariffs, and immigration restrictions.

The rest of the paper is organized as follows. Section 2 reviews the rapidly-expanding related literature. Section 3 provides an overview of the data and measurement. Section 4 lays out the empirical strategy. Section 5 presents the results. Section 6 concludes.

2 Related literature

Our paper is related to the growing literature that seeks to explain inflation dynamics in the post-pandemic period. This body of work has emphasized the importance of shocks to import prices and supply chains, with a natural focus on the goods sector due to its higher tradeability. For instance, Amiti et al. (2024) show that changes in import competition, along with a reduced ability to substitute between labor and intermediate inputs, contributed to the recent surge in inflation. Similarly, LaBelle and Santacreu (2022) highlight the significant impact of global supply chain disruptions on the U.S. Producer Price Index (PPI).

A separate strand of literature highlights the impact of tight labor markets on price pressures. Ball, Leigh and Mishra (2022) find that the V/U observed in 2021-22 can explain a significant portion of the rise in monthly core inflation. They note that "the contribution of V/U to the rise in 12-month inflation is 2.0 percentage points, nearly a third of the total inflation increase. However, the rise in V/U explains more—nearly one-half—of the rise in core inflation and, the effect of V/U is rising over time." Dao et al. (2024) also find that tight labor markets continue to contribute significantly to inflation in the United States, making it an exception among a large sample of advanced and emerging market economies, where inflationary pressure from relative price shocks has subsided, resulting in lower inflation. They find an even larger quantitative role for V/U in explaining price pressures in the United States compared to Blanchard and Bernanke (2024). This difference stems from their inclusion of V/U in their equation for core inflation, whereas in Blanchard and Bernanke (2024) V/U enters through the wage equation rather than the core-inflation equation directly. Our paper distinguishes itself from this literature by estimating both the link between labor market tightness and wages and the wage-price transmission using cross-sectional data and focusing on the services sector.

Unlike the aforementioned papers which use V/U, Crump et al. (2024) rely on the unemployment rate as their measure of labor market tightness and complement it with multiple measures of labor compensation. They project underlying inflation to remain high due to tight labor markets and strong wage growth.[4] Consistent with these papers, we find a strong role of the labor market in driving recent inflation. A key difference between our approach and these existing studies is that they examine time-series variation in aggregate inflation, while we explore cross-sectional variation across local markets. We provide a detailed comparison of the estimates in Section 5.5.

[4]Underlying inflation represents the inflation component that solely depends on the long-run trend and the sequence of current and future unemployment gaps.

Evidence on the nonlinearity of the Phillips curve is mixed. Benigno and Eggertsson (2023) show how nonlinearity can arise from a New Keynesian model with search and matching frictions and wage rigidity. Both Benigno and Eggertsson (2023) and Gitti (2024) provide empirical support for this nonlinear relationship, finding that inflation accelerates rapidly once V/U exceeds unity. However, Beaudry, Hou and Portier (2026) challenge the robustness of these findings. They show that the steepening at high V/U is largely attributable to unobserved inflation expectations rather than structural nonlinearity. In city-level panels, the nonlinearity disappears once time fixed effects or local household expectations are controlled for; in aggregate time series it survives only when expectations are proxied by expert forecasts (e.g., the Survey of Professional Forecasters) rather than by household or firm expectations. Evidence for Phillips curve nonlinearity therefore appears highly sensitive to how inflation expectations are measured and controlled for. We bring new evidence to this debate by targeting a different margin of the Phillips curve: rather than the reduced-form relationship between tightness and inflation, we focus on the pass-through from wage growth to services inflation. Despite including the same MSA and time fixed effects that Beaudry, Hou and Portier (2026) show are sufficient to eliminate nonlinearity in reduced-form V/U-inflation regressions, and despite additionally controlling for local inflation expectations, our estimated wage-price pass-through remains larger in the high V/U subsample than in the low V/U subsample (0.822 versus 0.594 at the 10-month horizon; the difference is 0.228 with a standard error of 0.025). Moreover, when we re-estimate our specification using the preferred dependent variables Beaudry, Hou and Portier (2026), headline and core inflation (excluding food and energy), in place of services inflation, we similarly find no evidence of nonlinearity. This helps reconcile the two sets of results: the amplification under tight labor markets appears to arise primarily because wage growth is passed through more strongly into service prices, rather than because tightness generates disproportionately larger wage increases. This discussion also connects to the broader literature that uses cross-sectional variation to identify the Phillips curve, as Beaudry et al. (2026) use a cross-sectional panel to show that conclusions about nonlinearities depend critically on whether common factors are controlled for.[5]

More broadlly, a growing strand of the literature estimates the Phillips curve using cross-sectional data (e.g., Beraja, Hurst and Ospina, 2019; McLeay and Tenreyro, 2019; Hooper, Mishkin and Sufi, 2020). Hazell et al. (2022) show theoretically that using regional data helps

[5]Our approach differs from Beaudry, Hou and Portier (2026). They estimate a reduced-form relationship between realized local labor market tightness and inflation, controlling for common factors via time fixed effects or direct measures of inflation expectations from surveys. We instead estimate a two-stage relationship: from tightness to service-sector wage growth, and from wage growth to services inflation, in an LP-IV framework in which wage growth is instrumented by a Bartik shock built from national tradable-sector tightness interacted with local exposure. This shift-share design targets a different source of endogeneity: rather than netting out unobserved aggregate expectations directly, it isolates plausibly exogenous cross-sectional variation in local demand, following the logic of Hazell et al. (2022). Notably, our baseline specification already includes MSA and time fixed effects, the same combination which Beaudry, Hou and Portier (2026) show is sufficient to eliminate spurious nonlinearity in reduced-form V/U-inflation regressions. Our vacancy measure also differs empirically: Beaudry, Hou and Portier (2026) apportion state-level JOLTS vacancies to MSAs using Census population weights, whereas we use MSA-month-level vacancy postings, benchmarked to JOLTS using the procedure described in Appendix A.3, which capture actual within-state cross-MSA variation.

overcome the issue of shifting long-run inflation expectations, which can confound the effect of labor market tightness. A cross-sectional framework also helps control for supply-side effect associated with monetary policy, as the central bank cannot offset regional demand shocks with a single national interest rate. Most estimations in this literature focus on the pre-pandemic period. For example, Hazell et al. (2022) estimate the slope of the regional Phillips curve using data across US states spanning 1978-2018. An exception is Barnichon and Shapiro (2024), who evaluate the performance of various tightness measures in predicting and explaining core inflation using MSA-level data at the quarterly frequency from 1982 to 2022. They find that V/U and vacancy filling cost proxies outperform other labor market tightness measures, such as the unemployment rate. Although both our paper and Barnichon and Shapiro (2024) estimate a cross-MSA Phillips curve, the focus and methodology differ substantially. Their analysis focuses on the direct relationship between various measures of labor market tightness and inflation. In contrast, we focus specifically on the wage-price pass-through, using microdata from the services sector. Additionally, our analysis is conducted at the monthly frequency, providing a more granular view than their quarterly analysis.

Taken together, the literature has established that both price shocks and labor market tightness contributed to the post-pandemic inflation surge, that the Phillips curve may be nonlinear, and that regional variation offers a promising setting for identification. We provide the first integrated evidence on the mechanism connecting these findings. Using granular, high-frequency microdata and a novel identification strategy, we trace the full transmission from local labor market tightness to wages to services inflation, and show that the resulting nonlinearity is concentrated specifically in the services sector.

3 Data and measurement

Our wage data come from a proprietary dataset provided by Homebase, a software company that offers scheduling, payroll reporting, and related services to businesses, primarily in retail, hospitality, and other service sectors. The dataset is based on timecard records, offering detailed information on work hours and wages from nearly 9 million workers across 1 million U.S. firms. It includes granular daily employee-level data on hours worked, wages, job types, and links to the corresponding establishments and parent firms. Additionally, establishment-level details include location (via 5-digit zip codes) and industry classification (via 6-digit NAICS codes). Our sample period spans from January 2019 to December 2022.[6] We calculate hourly wage (total wages divided by total hours) from worker-month records and then averaged across each MSA-month, with the 1st and 99th percentile tails trimmed.

Homebase distinguishes itself from conventional datasets with its granularity and frequency. For example, the Current Population Survey (CPS) provides detailed information on individual wages but does not provide sufficiently precise monthly measures at the MSA-by-service-sector level. The Current Employment Statistics (CES) provides high-frequency wage measures but has limited geographic and sectoral detail. The Quarterly Census of Employment and Wages (QCEW) provides local-industry employment and earnings information, but only quarterly frequency. Homebase offers a key advantage over these alternative data sources: it provides monthly wage information at the MSA level for service-sector

[6]The sample on prices and wages starts a year earlier to allow for the calculation of annual changes.

workers, enabling us to construct the wage measure required for our identification strategy. Nevertheless, Homebase has limitations. The sample concentrates on labor-intensive consumer services and it lacks data on tips, benefits, and overtime payments. In addition, the sample of firms varies over time, as Homebase adoption among businesses evolves. Despite these limitations, we show through external validity analyses, discussed in detail in Section 5.6, that the wage and employment trends in the Homebase sample closely track official statistics from the CPS and CES. Additionally, changes in employment and earnings align well with CES data at the month-state level (Dvorkin and Isaacson, 2022; Chen and Lee, 2024). Further, when replicating our analysis using official data instead of the Homebase sample—albeit with simplier specifications, lower-frequency data, and less granularity—the result qualitatively corroborates our baseline mechanism, suggesting that our conclusions may apply more broadly to the U.S. services sector.

For price data, we use CPI for 21 MSAs, sourced from the Bureau of Labor Statistics (BLS), which offers monthly or bi-monthly data at the MSA level. For MSAs with bimonthly CPI data, we interpolate linearly to obtain a monthly frequency.[7] Using these data, we construct year-on-year inflation series at the MSA-month level. Our baseline measure is services inflation excluding housing, and we also examine whether our main conclusions extend to headline and core inflation. By focusing on services, we primarily capture non-tradable, as nearly all services are non-tradable (Johnson, 2017).[8] We exclude housing inflation from our analysis because its drivers likely differ from those affecting non-housing services.[9] The literature has found that housing inflation exhibits a substantially different slope in the Phillips curve compared to non-housing services (Hazell et al., 2022; Stock and Watson, 2020), and that remote work has been a significant driver of post-pandemic housing prices (Howard, Liebersohn and Ozimek, 2023).

Table 1 presents the summary statistics of the sample. A first look at the data reveals an interesting pattern between services inflation and service sector wage growth at the aggregate level (Figure 3). Since mid-2021, these two series have shown a strong correlation with a lag, closely aligning with the timing of the rise in headline inflation.[10]

[7]For MSAs with bimonthly CPI data, the estimated 1-month ahead response may partly reflect interpolation rather than observed monthly price changes, because an interpolated value incorporates information from the next observed value.

[8]According to Johnson (2017), the only three CPI item codes for services classified as tradable are RA04 (Video cassettes, discs, and other media, including rentals), TF09 (unsampled motor vehicle fees), and TG01 (airline fare). Hazell et al. (2022) construct a non-tradable inflation series using BLS microdata, defining non-tradables similarly to the BLS service aggregation but with two exceptions: they classify Food Away from Home as non-tradable and exclude some transportation items (e.g., airline fare).

[9]We use data on CPI item "services less rent of shelter".

[10]The spikes in the wage growth series in early 2020 and early 2021 are consistent with aggregate evidence from the CES (Stewart, 2022) and reflect changes in workforce composition. At the onset of the pandemic, the disproportionate layoff of low-paid workers raised average hourly wages. This composition effect reversed in early 2021, as the economy reopened and employers refilled many of these positions. Time fixed effects in our regressions absorb common wage movements. Nevertheless, to the extent that our shift-share labor market tightness measure affects workforce composition differently across MSAs; our estimates may also capture changes in employment composition.

4 Empirical approach

We analyze the data using an LP-IV approach, as used in Jordà, Schularick and Taylor (2015) and Jordà, Schularick and Taylor (2020). This approach estimates impulse responses to shocks using two-stage least squares. Specifically, we estimate:

$$y_{i,t}^h = \alpha_t^h + \eta_i^h + \beta^h w_{i,t} + \sum_{k=1}^{K} \gamma_k^h w_{i,t-k} + \sum_{k=1}^{K} \delta_k^h y_{i,t-k} + \gamma^h X_{i,s,t} + u_{i,t}^h \tag{1}$$

where i, s, and t denote MSA, state, and month, respectively. $h = 0, 1, \ldots, H$ denotes the baseline estimation horizon. We control for lags up to $K = 3$. α_t^h is a time fixed effect. η_i^h is an MSA fixed effect. Our inflation measure, $y_{i,t}$, is year-on-year services inflation excluding housing in MSA i in month t calculated as the logarithmic difference in the prices for services excluding housing, $ln(prices)_{i,t} - ln(prices)_{i,t-12}$. The dependent variable $y_{i,t}^h \equiv ln(prices)_{i,t+h} - ln(prices)_{i,t+h-12}$, is year-on-year inflation h months ahead. $w_{i,t}$ is our independent variable of interest. It is year-on-year wage growth in the services sector from Homebase calculated as the logarithmic difference, $ln(wages)_{i,t} - ln(wages)_{i,t-12}$. We construct the wage measure as described in Section 3. Since inflation is measured as a 12-month change, we cluster standard errors $u_{i,t}^h$ at the MSA level to allow for within-MSA serial dependence (Jordà, 2005).

We include a vector of controls $X_{i,s,t}$. These include labor productivity from BLS as a proxy for local labor demand, measured at the state-year level. The controls also include a headline inflation shock defined as the difference between year-on-year headline inflation and core inflation (headline excluding food and energy) and calculated at the MSA-month level. This is intended to capture large price changes in particular sectors of the economy that are especially prominent during periods of supply-chain disruptions (Ball et al., 2022). Finally, residuals from regressions of previous horizons (i.e., from 0 to $h-1$) are also controlled. According to Teulings and Zubanov (2014), these residuals account for information between time t and $t+h$ that is not fully controlled by the other independent variables (which are as of time t).

The coefficient β^h captures the impulse response of inflation to wage growth over horizon h. We refer to this coefficient as the wage-price pass-through. Empirically, wages are typically less flexible than prices, making it unlikely that they will quickly adjust to changes in inflation. This is also evident in our sample period. As Figure 3 shows, the rise in wages precedes the rise in inflation.[11]

We instrument wage growth with a Bartik shock, following the intuition of Bartik (1991) and more recently Hazell et al. (2022).[12] In a nutshell, the Bartik shock is a shift-share instrument constructed based on national industry-level labor market tightness in the tradable sector (shift) and the employment share of those tradable industries within each MSA (share).

[11] The lagged response of inflation to wage growth is a common identification assumption in structural vector autoregression (SVAR) estimations. See, for example, Bernanke and Blanchard (2025).

[12] See Soh et al. (2022) and Chen and Lee, 2024 for applications of this method in the context of the pandemic.

A distinctive feature in our construction of the shift variable is that we use national changes in labor market tightness in the tradable sector only. Because prices in the tradable sector are set in national or global markets, our shift measure is unlikely to be directly correlated with unobserved shocks to wage growth in local non-tradable (service) sectors. In other words, we assume that national variations in labor market tightness in the tradable sector are driven by shocks that are specific to the tradable sector and uncorrelated with omitted variables that may affect local service sector wage growth. The intuition is that as positive (negative) shocks in the tradable sector increase (decrease) the demand for local non-tradables, firms in the non-tradable sector will experience an increase (decrease) in competition for workers, which ought to result in them raising (dropping) wages and prices. We provide a detailed discussion on our identification assumption below, and present robustness results when those assumptions may be violated in Section 5.

Our primary measure of labor market tightness is V/U (i.e., the number of vacancies divided by the number of unemployed workers), following recent studies (e.g., Ball, Leigh and Mishra, 2022; Blanchard, Domash and Summers, 2022; Benigno and Eggertsson, 2023). Several researchers have argued that V/U better reflects labor market tightness in the post-pandemic economy given the upward shift in the Beveridge curve post-pandemic (Furman and Powell, 2021, Barnichon and Shapiro, 2024, and Domash and Summers, 2022). The Beveridge curve—the relationship between vacancies and unemployment—has shifted outward, weakening the link between the unemployment rate and labor market tightness. This shift may also destabilize the traditional unemployment-based Phillips curve, with a given unemployment rate now associated with greater tightness and potentially higher inflation. The V/U better therefore provides a more direct measure of tightness in this environment. That said, some researchers argue that vacancies may be endogenous to inflation or may reflect changes in worker preferences towards more flexible work arrangements rather than labor market tightness (Afrouzi et al., 2024; Bagga et al., 2025; Guerreiro et al., 2024; and Pilossoph and Ryngaert, 2024). To address these concerns, we also construct alternative instruments using unemployment rate, employment growth, and wage growth.

Formally, the Bartik shock based on V/U in MSA i in month t is defined as:

$$Shock_{i,t} = \widehat{ln(V)}_{i,t} - \widehat{ln(U)}_{i,t}, \tag{2}$$

where $\widehat{ln(V)}_{i,t}$ is calculated as

$$\widehat{ln(V)}_{i,t} = \sum_{j=1}^{J} \phi_{i,j,2015-2016} * \ln(V_{j,t}). \tag{3}$$

$\ln(V_{j,t})$ is the log number of vacancies for 3-digit NAICS industry j in time t. $\phi_{i,j,2015-2016}$ is the average employment share of industry j in MSA i in 2015-2016. The set of industries is restricted to those classified as tradable sector, following the definition in Hazell et al. (2022). $\widehat{ln(U)}_{i,t}$ is constructed similarly.

We use industry-level data on unemployment from the CPS and vacancy data from a proprietary dataset provided by Indeed. Indeed is a global search engine for job listings. It collects job postings from various sources, including job listing sites, employer career

sites, and applicant tracking systems. Duplicated listings are removed, so each job is shown only once, even if it appears on multiple platforms. Our final dataset includes 142 million job postings, covering 421 occupations based on ISCO-08 classifications across 2.9 million companies and 576 counties in the United States.

The advantage of using Indeed, as opposed to the Job Openings and Labor Turnover Survey (JOLTS) from the BLS, is its higher frequency and greater granularity in geography and industry coverage. Indeed's industry classifications allow us to capture shocks at the 3-digit NAICS level, compared to the 2-digit level available in JOLTS. More importantly, Indeed's granularity data enable us to compute V/U at the MSA-month level—a measure not available from JOLTS—and thereby investigate the nonlinearity in wage-price pass-through based on local labor market conditions. A potential concern with the Indeed dataset is that it may not be representative of all job vacancies, since some are not posted online. To address this concern, we rescale the Indeed vacancies into nationally representative units using a procedure descried in Appendix A.3.

Earlier in this section, we discuss the intuition of the Bartik shock. Here we discuss the identification assumption more formally. Following the formulation of Borusyak, Hull and Jaravel (2022), our identification assumption is that shocks to local labor market tightness are quasi-randomly assigned. Quasi-randomness means that the shocks are uncorrelated with relevant unobservables in expectations and that a shock-level law of large numbers applies. The exposure share are allowed to be endogenous. Several points warrant discussion.

First, we maintain the quasi-random shocks as an identifying assumption. Because our Bartik shock is calculated using national-level variations in labor market tightness in the tradable sector, the underlying assumption is that this variation is uncorrelated, in expectation, with omitted variables that may affect local service sector wage growth.

Second, when shocks are measured using equilibrium objects—such as our baseline measure V/U and alternative measures of unemployment, employment, or wage growth—the Bartik shock can be interpreted as a noisy estimate of the true shock under certain conditions. One such condition is when the shocks are spatially uncorrelated. Since our local markets are large MSAs that are geographically dispersed across the country, it is reasonable to assume that shocks to local labor market tightness are not highly spatially correlated. To mitigate concerns of potential spatial correlation, we additionally control for region×time fixed effects to account for common conditions for MSAs in the same regional. We also report results with a leave-one-out (LOO) shock estimator, which removes spatial correlation arising from each MSA's own contribution to the national shift variable.[13] As a preview of our results, we find that estimates from the LOO estimator closely align with those from the baseline, and controlling for regional conditions yields similar findings.

Third, the shock-level law of large numbers assumption requires that the instrument incorporates many sufficiently independent shocks, each with a sufficiently small average exposure. Intuitively, this means relying on shocks from many industries relative to the number of local markets in the estimation. To support this assumption, our shocks are calculated at the 3-digit NAICS industry level, which provides a larger number of industry-

[13]In fact, the conventional non-LOO estimator is asymptotically valid in settings where many regions contribute to the shock calculation, which holds in our setting. Intuitively, this requires that a typical industry in the Bartik shock calculation is in a much larger number of regions than the number of industries that a typical region specializes in.

level shocks relative to the number of MSAs in our sample.

5 Results

5.1 Baseline

Figure 4 presents the baseline estimates for equation (1). The results provide evidence on two distinct margins of the relationship between labor market tightness and services inflation. The first stage estimates are positive and statistically significant at all horizons, indicating that tight local labor markets are associated with stronger wage growth in the services sector. The second stage estimates are positive and statistically significant at horizons from 2 to 12 months, indicating that higher service-sector wage growth is associated with subsequently higher services inflation. Together, the two stages are consistent with local labor market tightness transmitting to services inflation through wages.

The results are economically significant along both margins. For the tightness-to-wage margin, interpreting the logarithmic difference as a growth rate, the first stage estimate of 13.7 in the 3-month horizon implies that a one-unit increase in the shift-share tightness index is associated with a 13.7-percentage-point increase in the year-on-year wage growth rate.[14] For the wage-price margin, the second stage estimate of 30.3 percent over the 3-month horizon suggests that a one-percentage-point increase in year-on-year wage growth is associated with a 0.303-percentage-point increase in year-on-year inflation. This estimated pass-through increases over time, reaching 0.484 percentage points in the 10-month horizon. The two estimates can be combined to quantify the overall transmission from labor market tightness to services inflation through wages. At the 3-month horizon, a one-unit increase in the shift-share index implies an increase in year-on-year inflation of 4.1 percentage points through the wage channel. This effect rises to 6.0 percentage points at the 10-month horizon, reflecting the horizon-specific first-stage coefficient and wage-price pass-through.

Taken together, the two-stage estimates are consistent with two quantitatively important margins in the transmission of labor market tightness to inflation: tight labor markets generate large wage pressures, and a substantial, albeit not complete, share of those wage pressures is subsequently passed through to services prices. These implications differs from those of a traditional wage Phillips curve, which captures only the first part of the transmission from labor market conditions to inflation. Our results show that the second margin—wage-price pass-through—is critical for quantifying the overall inflationary effect of labor market tightness.

We next use the baseline estimates to assess the quantitative importance of these transmission mechanisms during the post-pandemic inflation episode. Figure 5 decomposes inflation into components associated with local labor market tightness and other factors, using the 3-month baseline estimates averaged across MSAs.[15] Local labor market tightness emerged as an important driver of inflation since mid-2021. It accounted for, on average, 34.2 percent of observed services inflation excluding housing between 2021Q3 and 2023Q1. This

[14]Because the instrument is a shift-share index, the coefficient is interpreted as the effect of a one-unit increase in the index rather than realized local V/U.

[15]The decomposition is based on the additive decomposition of fitted inflation.

contrasts with the period before mid-2021, when common shocks was an important driver, largely offsetting the effects of local labor market tightness.

In estimates for individual MSAs, the contribution of local labor market tightness is also evident as shown in Figure 6. Although the timing and magnitude of the inflation surge vary across MSAs, the model's fitted value closely tracks actual inflation in each MSA, accounting for between 20.4 and 33.9 percent of serivices inflation during 2021-2023, with an average of 26.3 percent (Figure 7).

5.2 Inflation measure comparison

The baseline results highlight an important role for wage pass-through in services inflation. This finding is consistent with a labor-cost channel through which local wage pressures are transmitted primarily to prices in labor-intensive, non-tradable services. If this mechanism is operative, wage-price pass-through should be stronger for services inflation than for broader price aggregates. To investigate this, we re-estimate our baseline specification using headline inflation and core inflation excluding food and energy as alternative dependent variables. Table 2 summarizes the results.

The contrast with our baseline estimates is quite telling. Unlike the substantial and persistent wage-price pass-through estimated for services inflation, the corresponding estimates of wage-price pass-through are negative when the dependent variable is either headline inflation or core inflation excluding food and energy. The negative coefficients may reflect offsetting movements in its non-services components as relative prices adjust. The substantial and persistent wage-price pass-through document in our baseline results is therefore specific to services inflation and is not apparent in broader inflation aggregates.

This comparison is important for interpreting the role of labor market conditions in inflation. A weak or negative relationship between local wage pressures and aggregate inflation need not imply that the wage channel is economically unimportant. Rather, aggregating price components with markedly different exposures to local labor costs can obscure a strong transmission mechanism within labor-intensive, non-tradable services. More broadly, our results show that conclusions about the inflationary effects of labor market tightness depend critically on the sectoral composition of the inflation measure being analyzed

5.3 Nonlinearity

Having established the two margins in the baseline specification, we next ask whether the strength of the wage-price transmission varies with local labor market conditions. This analysis contributes to the recent literature on Phillips curve nonlinearity by distinguishing between two potential sources of nonlinearity: the response of wages to labor market tightness and the subsequent transmission of wage pressures to prices. This distinction matters because a nonlinear relationship between labor market tightness and inflation may originate at either stage of the transmission mechanism. Recent theoretical work by Benigno and Eggertsson (2023) emphasizes the importance of nonlinearity in Phillips curves, but empirical evidence is mixed: some papers find evidence in favor of nonlinearities in the post-COVID period (Ball, Leigh and Mishra, 2022; Dao et al., 2024), while more recent work argues that this evidence

is fragile and may be driven by inadequate controls for inflation expectations (Beaudry, Hou and Portier, 2026).

Following Benigno and Eggertsson (2023), we define a labor shortage as a period in which V/U exceeds 1, indicating more job vacancies than workers seeking employment. Figure 8 shows the estimates separately for MSAs experiencing a labor shortage (High V/U) and those with V/U below one (Low V/U).[16] The first-stage estimates are similar across both subsamples, indicating that the response of service-sector wages to labor market tightness does not vary substantially with the degree of labor shortage. In contrast, the second-stage estimates are considerably larger in the high V/U subsample, particularly over the 6- to 12-month horizon. Thus, the stronger response of services inflation in unusually tight labor markets appears to arise primarily from greater wage-price pass-through rather than from a disproportionately larger response of wages to labor market tightness. One possible explanation is that, in exceptionally tight labor markets, persistent wage pressures exhaust firms' ability to absorb higher labor costs through profit margins or productivity, leading to greater pass-through to services prices.

This finding is relevant to the recent debate over the role of inflation expectations in generating Phillips curve nonlinearities. Beaudry, Hou and Portier (2026) show that controlling for local inflation expectations, together with time fixed effects, can substantially attenuate estimates of nonlinearity. Our baseline specification already includes MSA and time fixed effects. We therefore follow Beaudry, Hou and Portier (2026) and additionally control for local inflation expectations.[17] Table 3 shows that our estimates are very similar with this additional control. Similarly, when we control for local inflation expectations in the high V/U and low V/U samples, the results are essentially unchanged. In our setting, therefore, the stronger wage-price transmission in unusually tight labor markets does not appear to be explained by differences in local inflation expectations.

We next ask whether the nonlinear wage-price transmission is specific to services or reflects a broader relationship with other prices. We re-estimate our specification using alternative dependent variables: headline inflation and standard measure of core inflation (headline excluding food and energy). In contrast to our services results, we find no evidence of a differential response across the high V/U and low V/U subsamples (Table 4). The nonlinear transmission we document is therefore concentrated in services inflation. This finding helps reconcile our results with Beaudry, Hou and Portier (2026): nonlinearity is concentrated in services and may be difficult to detect in broad price aggregates.

Taken together, these results offers new insights into where in the labor market-inflation transmission mechanism the nonlinearity arises. We find little evidence that unusually tight labor markets generate a disproportionately larger wage response. Rather, the amplification occurs at the subsequent wage-price margin: a given increase in service-sector wages is associated with a larger increase in services inflation when labor markets are unusually tight.

[16] As in the baseline, identification within each regime comes from the national shift-share instrument, rather than from the realized V/U used to define the split.

[17] We use inflation expectations from the Survey of Consumer Expectations (SCE), following Beaudry, Hou and Portier (2026). Because the SCE provides expectations at the commuting-zone level, we map the SCE commuting-zone expectations into our MSA panel (assigning each commuting zone to the sample MSA with the most mapped counties, with estimation and clustering remaining at the MSA level), first without controlling for local inflation expectations and then including them as an additional control.

Moreover, this state dependence is specific to services and is not evident in broader inflation aggregates. The results therefore suggest that the nonlinearities emphasized in the recent Phillips curve literature can reflect heterogeneity in the wage-price transmission mechanism, which may be obscured when inflation is measured using broad price aggregates.

5.4 Extension and robustness

Common shocks As emphasized by the recent literature on the Phillips curve (McLeay and Tenreyro, 2019, Hazell et al., 2022, Gitti, 2024), cross-sectional data are valuable for identifying the slope of the Phillips curve. In a similar vein, the cross-sectional dimension of our data is important for identifying the pass-through from wage growth to inflation, as it allows us to control for common shocks that affect both services inflation and wage growth. In our setting, the inclusion of month fixed effects means that the wage coefficient is identified from differences in wage growth across MSAs within a given month, thereby absorbing common cost push shocks and aggregate inflation expectations. This distinction is empirically important as we show in Table 5. At the 3-month horizon, the second-stage wage-price pass-through is 0.303 and statistically significant when month fixed effects are included, but falls to 0.057 when month fixed effects are omitted. At the 10-month horizon, the pass-through is 0.484 when month fixed effects are included, but falls to -0.025 when month fixed effects are omitted. The comparison shows sensitivity in the wage-price margin, while the first-stage coefficients remain broadly stable. Our results therefore highlight the value of exploiting cross-MSA variation to identify wage pass-through while holding common aggregate conditions fixed.

Shift-share instrument As discussed in Section 4, the identification assumption of our shift-share instrument requires spatially uncorrelated shocks. We now investigate the robustness of our results when this assumption might be violated. To address potential spacial correlation arising from the construction of the Bartik shock, we employ a leave-one-out (LOO) estimator (Borusyak et al., 2022).[18] Figure 9 shows that results from a LOO estimator are similar to the baseline. To further mitigate potential spatial correlation, we control for common conditions at the regional level using region×time fixed effects. Since our local markets are large MSAs that are geographically dispersed, spatial correlation is most likely to occur among MSAs within the same region. As shown in Figure 10, our results remain robust with these controls.

Our next test directly addresses potential confounding factors at the MSA level. If these factors are correlated with the national tradable-sector shocks underlying the Bartik measure and also directly affect on local service sector wage growth, the quasi-random shock assumption would be violated. We therefore additionally control for household income and housing wealth at the MSA level, as increases in household income and housing wealth during our sample period—possibly reflecting stimulus spending and rising housing prices—may affect local labor supply and service-sector wage growth. Figure 11 shows that our results are robust to these controls.

[18]The LOO estimator is calculated by leaving out observations from MSA i when calculating $ln(V_{j,t})$ and $ln(U_{j,t})$ for $Shock_{i,t}$.

As a point of comparison, Figure 12 presents results using an alternative Bartik shock constructed from national-level variation in labor market tightness across all sectors rather than tradable sectors alone. Because this measure does not isolate variation originating tradable sectors that is plausibly exogenous to local labor market conditions, we interpret the results as descriptive rather than causal. Relative to the baseline, the first-stage estimates are larger, in the range of 25–30 compared to 12-15 in the baseline. The second-stage estimates, however, are smaller, peaking at about 0.3 compared with 0.5 in the baseline.

Labor market measures Finally, we explore alternative measures of labor market tightness. In our baseline estimation, we use V/U following a recent literature, which accounts for the upward shift in the Beveridge curve in the post-pandemic economy. Here, we examine how results based on traditional measures of labor market tightness compare, also addressing the potential concern that vacancies may reflect factors other than labor market tightness. Specifically, we use the unemployment rate, employment growth, or log payroll levels as the shift variable to construct the Bartik shock. The results are shown in Figure 13 for the unemployment rate measure, Figure 14 for the employment growth measure, and Figure 15 for the log payroll levels measure. The first-stage estimates are qualitatively similar when using employment or log payroll levels, although the coefficients are larger, likely reflecting differences in the scale of these measures. The second-stage estimates are also broadly similar when using employment or log payroll levels, while the coefficient is larger when using the unemployment rate. One difference that stands out is the small and statistically insignificant first-stage relationship when using unemployment rate. Unlike V/U, the unemployment rate alone may not fully capture labor market tightness in the post-pandemic period, when the Beveridge curve shifted outward and elevated vacancy rates coexisted with relatively low unemployment. Thus, unemployment provides less information about the demand for labor and the resulting wage pressures than a measure such as V/U that incorporates both vacancies and unemployment.

5.5 Comparison with recent literature

Our contribution differs from most estimates of the Phillips curve in the empirical relationship we identify. Existing studies typically estimate either a wage Phillips curve, relating labor market tightness to wage growth, or a price Phillips curve, relating tightness directly to inflation. These two relationships provide important pieces of the transmission mechanism, but neither by itself identifies the link between labor market conditions, wages, and prices. Tight labor markets may generate substantial wage pressures without producing equivalent price pressures if firms absorb higher labor costs. Conversely, a reduced-form relationship between tightness and inflation does not establish that wages are the transmission channel. Our two-stage approach allows us to identify both margins separately. In this sense, our estimates provide a link between the wage Phillips curve and the price Phillips curve.

The sectoral focus of our analysis is important for identifying this transmission mechanism. As we show in Section 5.2, the substantial and persistent wage-price transmission that we identify differs from the results for headline and standard core inflation. Thus, the absence of a strong relationship between local wage pressures and headline inflation need not imply that the wage channel is unimportant. Our results therefore suggest that part of

the disagreement in the Phillips curve literature reflects differences not only in identification and measures of labor-market tightness, but also in the sectoral composition of the inflation measure.

With these distinctions in mind, this section compares our results with recent estimates of the relationship between labor market tightness and inflation. Ball, Leigh and Mishra (2022) estimate the relationship between log V/U and core inflation, measured by median inflation, and find evidence of nonlinearities with respect to labor market tightness. Their estimates for 2020-2021 imply that an increase in V/U from 0.5 to 1.5 corresponds to higher core inflation by approximately 3.5 percentage points. Benigno and Eggertsson (2023) estimate the relationship between log V/U to core inflation, measured as inflation excluding food and energy. Over 2008-2022, they estimate a one-quarter inflation response of 4.7 percentage points when V/U exceeds 1, and 0.5 percentage points without when V/U is below 1. By comparison, our two-stage estimates imply that a one-unit increase in tightness, as measured by the shift-share index, is associated with a 4.6 percentage points increase in services inflation excluding housing of over a 3-month horizon when V/U exceeds 1 and 2.8 percentage points when V/U is below 1. These magnitudes are not directly comparable, however, for two reasons. First, our estimates measure the inflation response to a one-unit increase in the shift-share tightness index rather than in realized V/U. Second, our baseline outcome is services inflation excluding housing, whereas the other studies examine core inflation.

Our second-stage estimates can also be compared with studies that directly examine the wage-price relationship. Using aggregate data from 1990Q1 to 2023Q2, Bernanke and Blanchard (2025) estimate a wage-price pass-through of about 0.7 for headline inflation, calculated as the sum of contemporaneous coefficient and coefficients on four quarterly lags. By comparison, our estimated pass-through to services inflation excluding housing peaks at 0.48 at the 10-month horizon, while the sum of the coefficients at quarterly horizon over 1 year is about 1.6. These estimates are not directly comparable because they differ in both the inflation measures and in the empirical objects being estimated. Their estimate captures the aggregate relationship between wage and headline inflation, including common movements in costs and inflation expectations. Our cross-sectional estimate, by contrast, is identified from differences across local labor markets within a given month while absorbing common aggregate movements. Our estimate should therefore be interpreted as capturing the local services component of the wage-price transmission rather than the aggregate wage-price relationship.

5.6 External validity

Our empirical strategy relies on Homebase because its combination of monthly frequency, MSA-level coverage, and service-sector detail provides the variation needed to identify the two-stage transmission from local labor market tightness to wages and from wages to services inflation. This raises an important question: to what extent do our findings reflect the underlying wage and inflation dynamics in the U.S. services sector rather than features of the Homebase sample or our specification? We address this question in three ways. First, we compare wage movements in Homebase with those in official statistics. Second, we replicate the two-stage analysis using official wage data at the quarterly MSA level. Third, we examine whether the broader relationship between labor market tightness and services inflation is also

evident in the aggregate time series.

First, we assess whether Homebase captures wage dynamics that are consistent with official statistics. At the national level, Figure 16 shows that the average earnings trend from the Homebase sample closely aligns with those from the CPS and the CES for a comparable services sector, consistent with the evidence in Dvorkin and Isaacson (2022) and Chen and Lee (2024). Figure 17 similarly shows that year-on-year hourly wage growth from Homebase closely tracks wage growth in the CES. The correspondence is also evident across states: monthly hourly wage and total earnings, indexed to January 2020, in Homebase are highly correlated with those in the CES (Figure 18). These comparisons indicate that the wage dynamics underlying our estimates are not driven by systematic differences between Homebase workers and workers covered by official statistics.

We next ask whether our two-stage results are robust to replacing the Homebase wage measure with lower-frequency official statistics. Because official wage data do not provide the monthly MSA-level variation required for our baseline specification, we use year-on-year growth in total payroll in the leisure and hospitality sector from the QCEW and re-estimate the two stages at the MSA-quarter level. The sample covers 2019Q1-2022Q4, consistent with the baseline. Figure 19 shows that the first-stage coefficient ranges from 30.2 to 43.6 over the 1- to 4-quarter horizon. The second-stage coefficient rises over the same horizon, ranging from 0.14 to 0.16. Combining these two stages yields an implied effect of labor market tightness on services inflation that peaks at 6.0 percentage points in the 3-quarter horizon. These magnitudes are broadly similar to our baseline estimates, although the coefficients are less precisely estimated and the payroll-to-price coefficient declines more rapidly. This exercise should be interpreted as a qualitative comparison, because total payroll captures changes in employment and hours as well as wages, whereas our baseline measure captures hourly wages.

The QCEW exercise also provides a useful test of the broader inflation implications of the mechanism. Figure 20 decomposes the estimation results at the 1-quarter horizon. As in the baseline analysis, the model closely tracks actual inflation, and local labor market tightness has emerged as the key driver since mid-2021. Thus, despite the lower frequency and greater aggregation of the official data, they reproduce the central qualitative finding of the baseline analysis: local labor market conditions generate wage pressures that are subsequently reflected in services prices. The comparison therefore supports the interpretation that our findings are not specific to the Homebase data. The granular Homebase data improve our ability to identify the mechanism rather than generate the mechanism itself.

Finally, we examine whether the relationship between labor market tightness and services inflation extends beyond the cross-sectional variation used in our baseline analysis. We estimate an aggregate Phillips curve in which national services inflation excluding housing depends on the log V/U. Details of the specification and data are provided in Appendix A.2, and key findings are summarized here. Figure 21 shows that the coefficient for labor market tightness is positive and statistically significant. The estimated response is 0.6 at the 3-month horizon and increases to between 1.5 and 2.0 at the 12-month horizon and beyond. Rolling regressions with a 3-year window reveal substantial variation in the strength of this relationship over time (Figure 22).[19]

[19]Benigno and Eggertsson (2023) report a coefficient of 0.6 in the full sample and 4.5 for the post-2008

The aggregate time-series estimates also illustrate the quantitative relevance of labor market conditions for post-pandemic evolution of services inflation. Figure 23 compares the fitted values from the rolling window regression with actual inflation. The specification incorporating labor market tightness closely tracks the sharp increase and subsequent decline in services inflation during 2021-2023. In contrast, a counterfactual specification that sets V/U to 1 throughout the sample predicts an earlier surge and a faster decline in inflation. The counterfactual fitted path's peak is 1 percentage point lower and falls below 2 percent by early 2023.[20] Together with the MSA-level decomposition in Figure 7, these results show that the explanatory power of labor market conditions is evident both across local labor markets and in the aggretate time series.

Taken together, these exercises provide evidence that our findings are not an artifact of either the Homebase sample or our two-stage, cross-sectional empirical strategy. Homebase wage movements closely track official statistics, the two-stage relationship between labor market tightness, wages, and services inflation is corroborated using quarterly QCEW data, and the broader relationship between labor market tightness and services inflation is also evident in the aggregate time series. At the same time, the comparison highlights why the Homebase data are important. Official data can corroborate the underlying relationship, but their frequency and geographic aggregation do not permit us to separately identify the two stages at the monthly MSA level. The granular Homebase data therefore do not merely provide an alternative measure of wages; they make it possible to trace the transmission from local labor market tightness to wage growth and, in turn, to services inflation with a degree of temporal and geographic resolution unavailable in standard data sources.

6 Conclusion

This paper examines the transmission mechanism linking labor-market conditions to inflation: the pass-through from labor-market-induced wage growth to services prices. Much of the Phillips curve literature estimates either how labor market tightness affects wages or how it affects aggregate inflation. These approaches are informative but leave open the question of whether wage pressures generated by tight labor markets are subsequently passed through into prices. Using cross-sectional variation across U.S. local labor markets and an LP-IV framework, we provide direct evidence on this missing link.

We find a strong transmission from local labor market tightness to service-sector wage growth and from service-sector wage growth to services inflation. Taken together, these results imply that prolonged labor market tightness can generate persistent inflation through the wage channel. The mechanism is economically important for services inflation but is attenuated when broader price aggregates are used. This finding helps reconcile why the estimated role of labor markets can differ across studies focusing on wage, or different measures of inflation: a strong wage-price mechanism within labor-intensive and locally priced services can be obscured when inflation measures include components driven primarily by

sample during periods of labor shortage.

[20]This is an in-sample decomposition. The fitted value from the counterfactural specification sets the contribution from current and three lagged log(V/U) to zero while retaining other controls. The contribution of labor market tightness is the gap between the two fitted paths.

tradables.

We also find that the transmission from wages to services prices becomes stronger when labor markets are unusually tight. This result suggests that Phillips curve nonlinearity operates primarily through greater wage-price pass-through when labor shortages emerge. Thus, identifying the wage-price link is also important for understanding the state dependence of inflationary pressures.

More broadly, our findings suggest that the relevant question for monetary policy is not simply whether tight labor markets affect inflation, but through which sectors and transmission margins their effects operate. During the post-pandemic inflation episode, supply chain disruptions and commodity-price shocks were central to the surge in goods inflation, while persistent wage pressures in the services sector contributed to inflation persistence after those shocks receded. Monitoring this channel is therefore important for assessing inflation persistence and the pace at with which inflation is likely to return to target.

	Obs	Mean	Std. Dev.	Min	Median	Max
Services inflation (excl. housing)	906	3.30	2.41	-3.18	2.87	10.03
Wage growth	906	0.95	17.22	-66.33	3.40	63.33
Bartik shock	906	-0.05	0.64	-2.05	0.02	0.83
Vacancy-to-unemployment ratio	906	1.07	0.54	0.15	1.04	3.53
Labor productivity	906	1.94	3.10	-4.50	2.10	7.70
Headline shock	906	0.54	1.03	-1.81	0.35	3.89
Log household income per capita	906	11.15	0.19	10.65	11.13	11.73
Log housing wealth	906	4.97	0.23	4.40	4.97	5.55

Table 1: Summary statistics

Notes: This table reports the summary statistics. The sample is a monthly panel of MSAs over January 2019 to December 2022. Observations are weighted by the average size of the labor force in each MSA in 2019. We measure inflation as year-on-year logarithmic difference in the price level, $ln(prices)_{i,t} - ln(prices)_{i,t-12}$ for MSA i and month t. Wage growth denotes year-on-year wage growth in the service sector from Homebase measured by logarithmic difference, $ln(wage)_{i,t} - ln(wage)_{i,t-12}$. We instrument wage growth using the Bartik shock defined in equation (2) and constructed using tradable-sector industries following the definition in Hazell et al. (2022). Labor productivity (from BLS) proxies for local labor demand. Headline inflation shocks are defined as the difference between year-on-year headline and core excluding food and energy inflation. Housing wealth is defined as the product between Census homeownership rates and the Freddie Mac House Price Index. Source: BLS, Homebase, and authors' calculations.

Dependent variable (inflation measure)	β^3	β^{10}
Services excluding housing	0.330^{***}	0.536^{***}
	(0.042)	(0.058)
Headline	-0.135^{***}	-0.137^{***}
	(0.041)	(0.029)
Core (excluding food and energy)	-0.266^{***}	-0.556^{***}
	(0.033)	(0.027)

Table 2: Wage-price pass-through by inflation measure

Notes: Each row re-estimates the baseline LP-IV (equation (1)) replacing the dependent variable with the indicated inflation measure. All rows omit the headline inflation shock control (the difference between year-on-year headline and core inflation), which mechanically contains the headline and core dependent variables; with this control, the services estimates are 0.303 and 0.484 (Figure 4). Statistical significance is accessed using weak-IV-robust (Anderson–Rubin) p-values. The first-stage Kleibergen-Paap F statistics at $h = 3$ and $h = 10$ are 5.14 and 4.00 (services), 2.41 and 2.46 (headline), and 3.29 and 2.70 (core). β^h is the second-stage wage-price pass-through coefficient at horizon h (in months); standard errors clustered by MSA in parentheses. ***, **, and * denote significance at the 1, 5, and 10 percent levels. Source: BLS, Homebase, and authors' calculations.

	Without expectations control	With expectations control
Horizon $h = 3$		
First-stage coefficient	16.99	17.98
Second-stage β^3	0.330^{***}	0.319^{***}
	(0.035)	(0.034)
Horizon $h = 10$		
First-stage coefficient	14.88	15.95
Second-stage β^{10}	0.661^{***}	0.611^{***}
	(0.046)	(0.041)

Table 3: Robustness to local inflation expectations

Notes: The baseline LP-IV is re-estimated on the MSA sample to which SCE commuting-zone inflation expectations are mapped (20 of 21 MSAs), first without and then with the MSA-level one-year-ahead SCE inflation expectation (and three lags) as an additional control. Standard errors clustered by MSA in parentheses. *** denotes significance at the 1 percent level. Source: BLS, Homebase, NY Fed SCE, and authors' calculations.

Inflation measure	High V/U	Low V/U	Difference
Services excluding housing	0.822***	0.594***	0.228***
	(0.054)	(0.040)	(0.025)
Core (excluding food and energy)	−0.333***	−0.296***	−0.038
	(0.023)	(0.020)	(0.018)
Headline	−0.082***	−0.001	−0.081**
	(0.020)	(0.022)	(0.013)

Table 4: Wage-price pass-through by inflation measure, high vs. low V/U

Notes: Second-stage wage-price pass-through (β^{10}) at the 10-month horizon from the baseline LP-IV, estimated jointly with the pass-through allowed to differ between MSA-months with a labor shortage (High V/U > 1) and without (Low V/U ≤ 1). The difference column reports the High−Low gap using the joint covariance. Statistical significance is accessed using weak-IV-robust (Anderson–Rubin) p-values. Estimates are on the SCE-covered subsample (20 of 21 MSAs) with local inflation expectations controlled, so the levels differ from the full-sample split (High 0.746 and Low 0.469 for services). Kleibergen-Paap rk Wald F statistics are 8.06, 1.77, and 4.30 for services excluding housing, core, and headline inflation, respectively. The core and headline rows omit the headline inflation shock control, which mechanically contains those dependent variables. Standard errors clustered by MSA in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

	With time FE (baseline)	Without time FE
Horizon $h = 3$		
First-stage coefficient	13.69	13.48
Second-stage β^3	0.303^{***}	0.057^{***}
	(0.039)	(0.004)
Horizon $h = 10$		
First-stage coefficient	12.47	13.57
Second-stage β^{10}	0.484^{***}	-0.025^{***}
	(0.062)	(0.006)

Table 5: The role of common shocks: with and without time fixed effects

Notes: The baseline specification (with time fixed effects) is compared to one omitting time fixed effects. Standard errors clustered by MSA in parentheses. *** denotes significance at the 1 percent level. Source: BLS, Homebase, and authors' calculations.

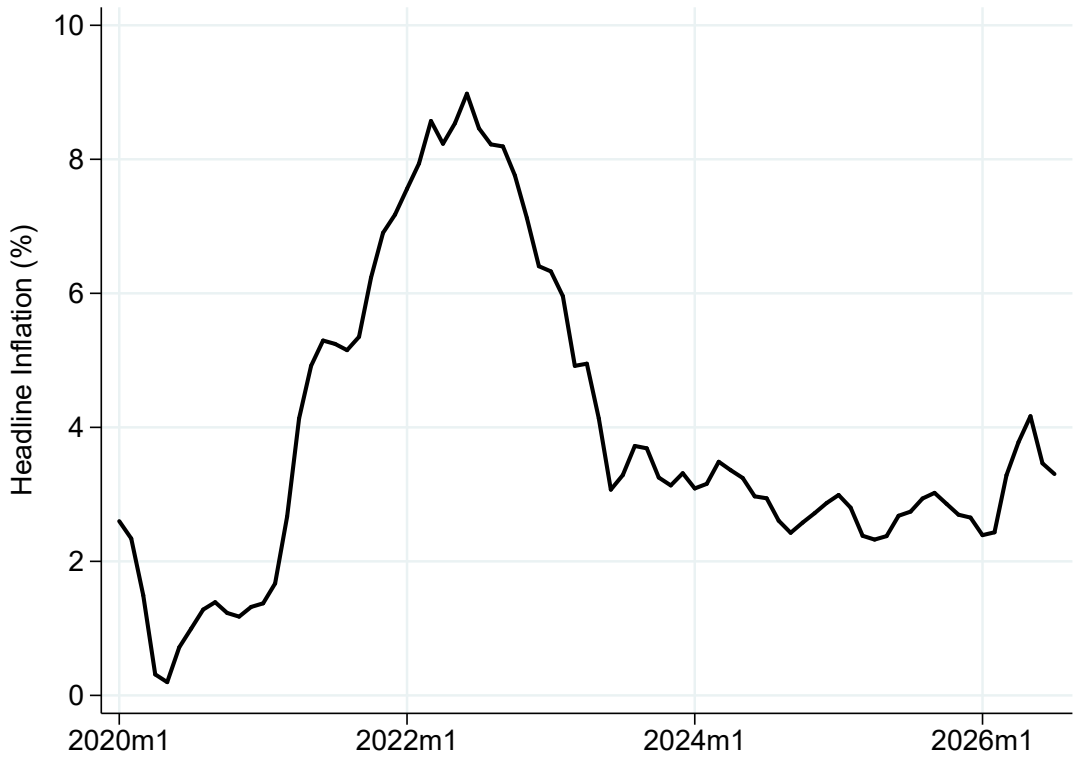


Figure 1: Headline inflation

Notes: This figure plots the year-on-year US headline inflation. Source: BLS and authors' calculations.

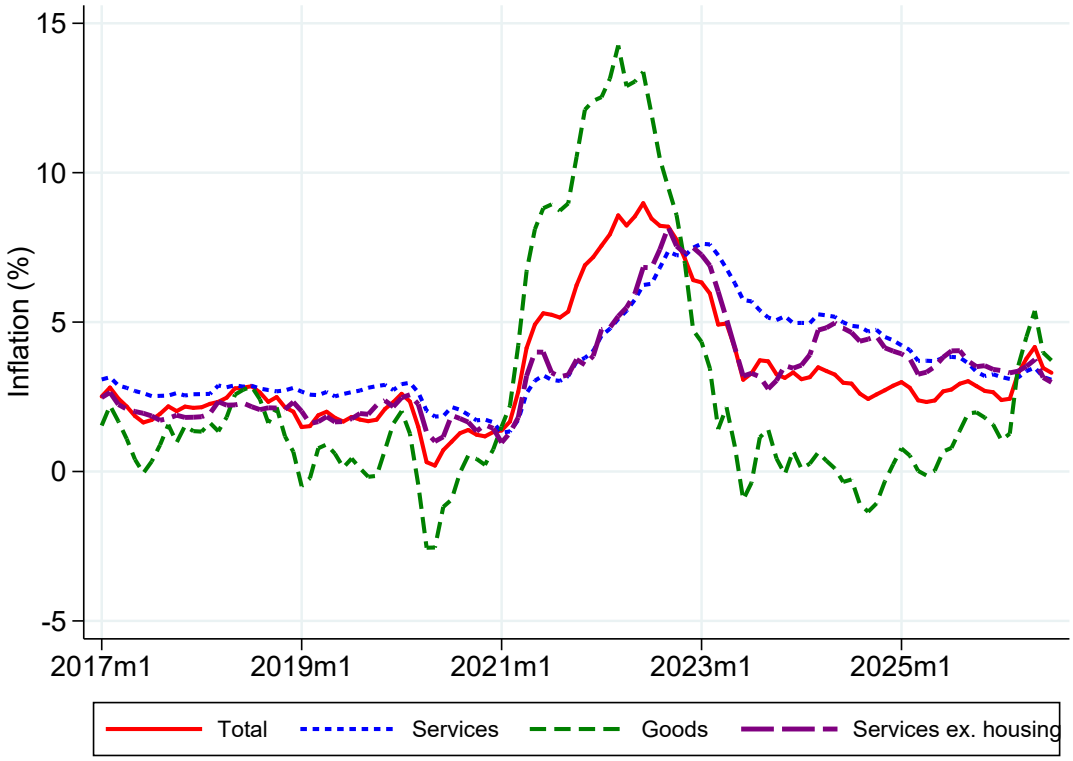


Figure 2: Headline, services, and goods inflation

Notes: This figure plots year-on-year CPI inflation for all items (Total), services, goods, and services excluding housing (services less rent of shelter), through July 2026. Source: BLS and authors' calculations.

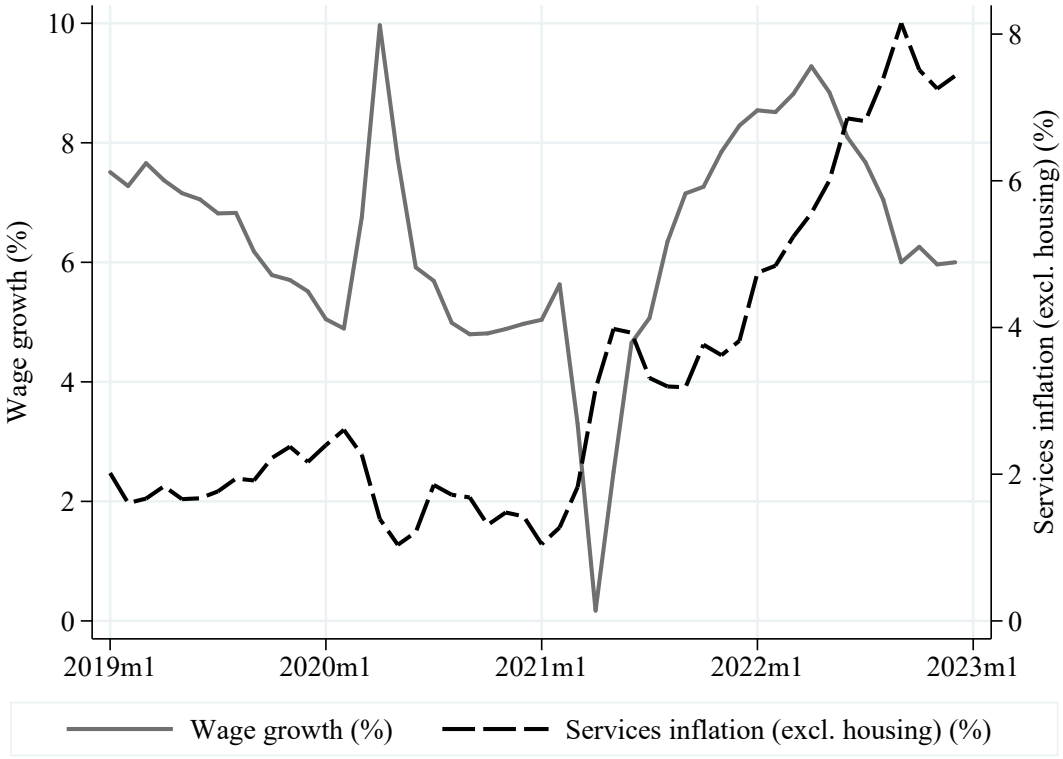


Figure 3: Price and wage inflation

Notes: This figure plots year-on-year average hourly wage growth from Homebase (left axis) and year-on-year services (excluding housing) inflation (right axis) from the BLS. Source: BLS, Homebase, and authors' calculations.

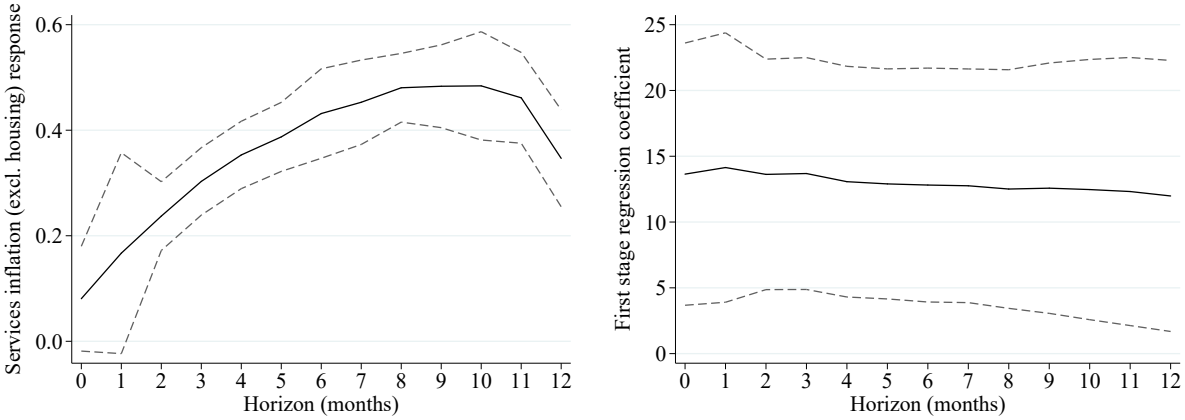


Figure 4: Services inflation (excluding housing) response, MSA level baseline model

Notes: This figure plots first and second stage estimates from the LP-IV model in equation (1):

$$y_{i,t}^h = \alpha_t^h + \eta_i^h + \beta^h w_{i,t} + \sum_{k=1}^{K} \gamma_k^h w_{i,t-k} + \sum_{k=1}^{K} \delta_k^h y_{i,t-k} + \gamma^h X_{i,s,t} + u_{i,t}^h$$

where i, s, and t denote MSA, state, and month respectively. $h = 0, 1, \ldots, H$ denotes the estimation horizon. We control for lags up to $K = 3$. $y_{i,t}$ is year-on-year inflation of MSA i in month t measured by the logarithmic difference in the prices for services excluding housing, $ln(prices)_{i,t} - ln(prices)_{i,t-12}$. The dependent variable $y_{i,t}^h$ denotes year-on-year inflation at $t + h$. α_t^h is a time fixed effect. η_i^h is an MSA fixed effect. $w_{i,t}$ is our independent variable of interest, year-on-year wage growth in the service sector from Homebase measured by logarithmic difference, $ln(wage)_{i,t} - ln(wage)_{i,t-12}$. We instrument wage growth using the Bartik shock defined in equation (2) and constructed using tradable-sector industries following the definition in Hazell et al. (2022). We also include a vector of controls $X_{i,s,t}$ at their native geographic and temporal levels, including labor productivity (from BLS) to proxy for local labor demand, headline inflation shocks defined as the difference between year-on-year headline and core excluding food and energy inflation, and residuals from the regressions for previous horizons. The sample is a monthly panel of MSAs over January 2019 to December 2022. Observations are weighted by the average size of the labor force in each MSA in 2019. We cluster the standard errors $u_{i,t}^h$ at the MSA level. The solid line shows the point estimates and the dashed lines show the 90 percent confidence interval. Source: BLS, Homebase, and authors' calculations. The first-stage coefficient concerns contemporaneous growth for each horizon-specific estimation sample and controls, not growth accumulated over the horizon.

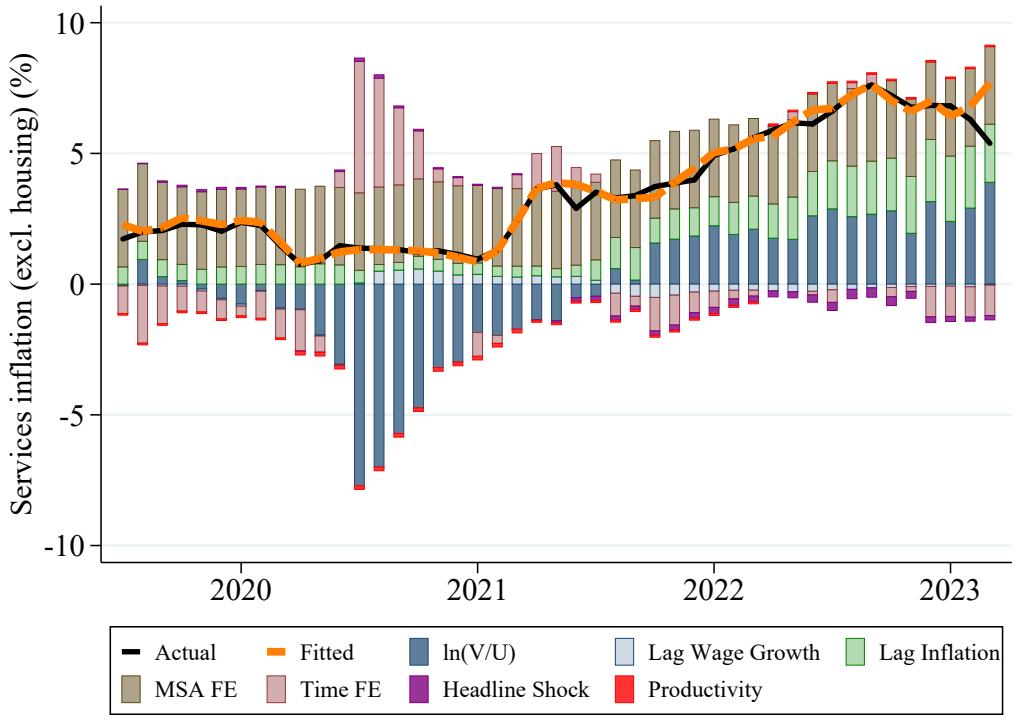


Figure 5: Predictions for services inflation (excluding housing), MSA level baseline model

Notes: This figure plots fitted values from the LP-IV model estimated at the MSA level (Figure 4):

$$y_{i,t}^h = \alpha_{2,t}^h + \eta_{2,i}^h + \beta_2^h \widehat{w}_{i,t} + \sum_{k=1}^{K} \gamma_{2,k}^h w_{i,t-k} + \sum_{k=1}^{K} \delta_{2,k}^h y_{i,t-k} + \gamma_2^h X_{i,s,t} + u_{2,i,t}^h$$

$$w_{i,t} = \alpha_{1,t}^h + \eta_{1,i}^h + \beta_1^h Shock_{i,t} + \sum_{k=1}^{K} \gamma_{1,k}^h w_{i,t-k} + \sum_{k=1}^{K} \delta_{1,k}^h y_{i,t-k} + \gamma_1^h X_{i,s,t} + u_{1,i,t}^h$$

where i, s, and t denote MSA, state, and month respectively. $h = 3$ denotes the estimation horizon. All values are unweighted averages across MSAs in the sample. The solid line plots actual inflation, and the dashed line plots fitted values from the LP-IV model. The bars show the contribution of each independent variable in the LP-IV model, where the contribution captures the combined effect of each variable from the first and second stages. "ln(V/U)" plots the fitted value from the Bartik shock $Shock_{i,t}$ for the log vacancy-to-unemployment ratio defined in equation (2), constructed using only tradable-sector industries following the definition in Hazell et al. (2022). "Lag Wage Growth" contains fitted values from $w_{i,t-k}$, "Lag Inflation" from $y_{i,t-k}$, "MSA FE" from $\eta_{1,i}^h$ and $\eta_{2,i}^h$, "Time FE" from $\alpha_{1,t}^h$ and $\alpha_{2,t}^h$, and "Headline Shock" and "Productivity" from the controls in $X_{i,s,t}$. The 34.2 percent figure reported in the text is the average "ln(V/U)" contribution as a share of average observed services-excluding-housing inflation across MSAs over Q3 2021–Q1 2023, a descriptive attribution from this additive decomposition rather than an identified causal share. Source: BLS, Homebase, and authors' calculations.

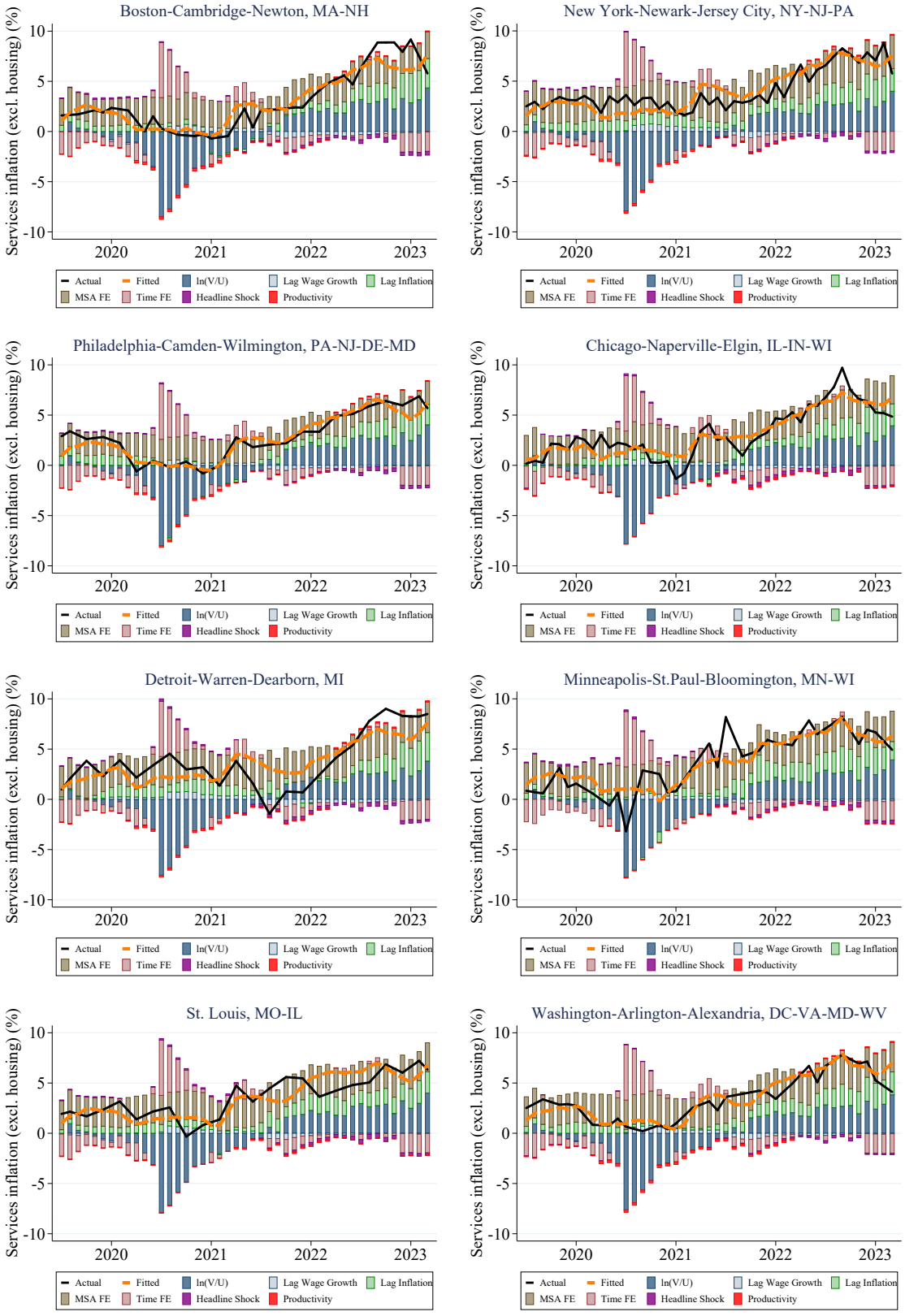


Figure 6: Predictions for services inflation (excluding housing) by MSA

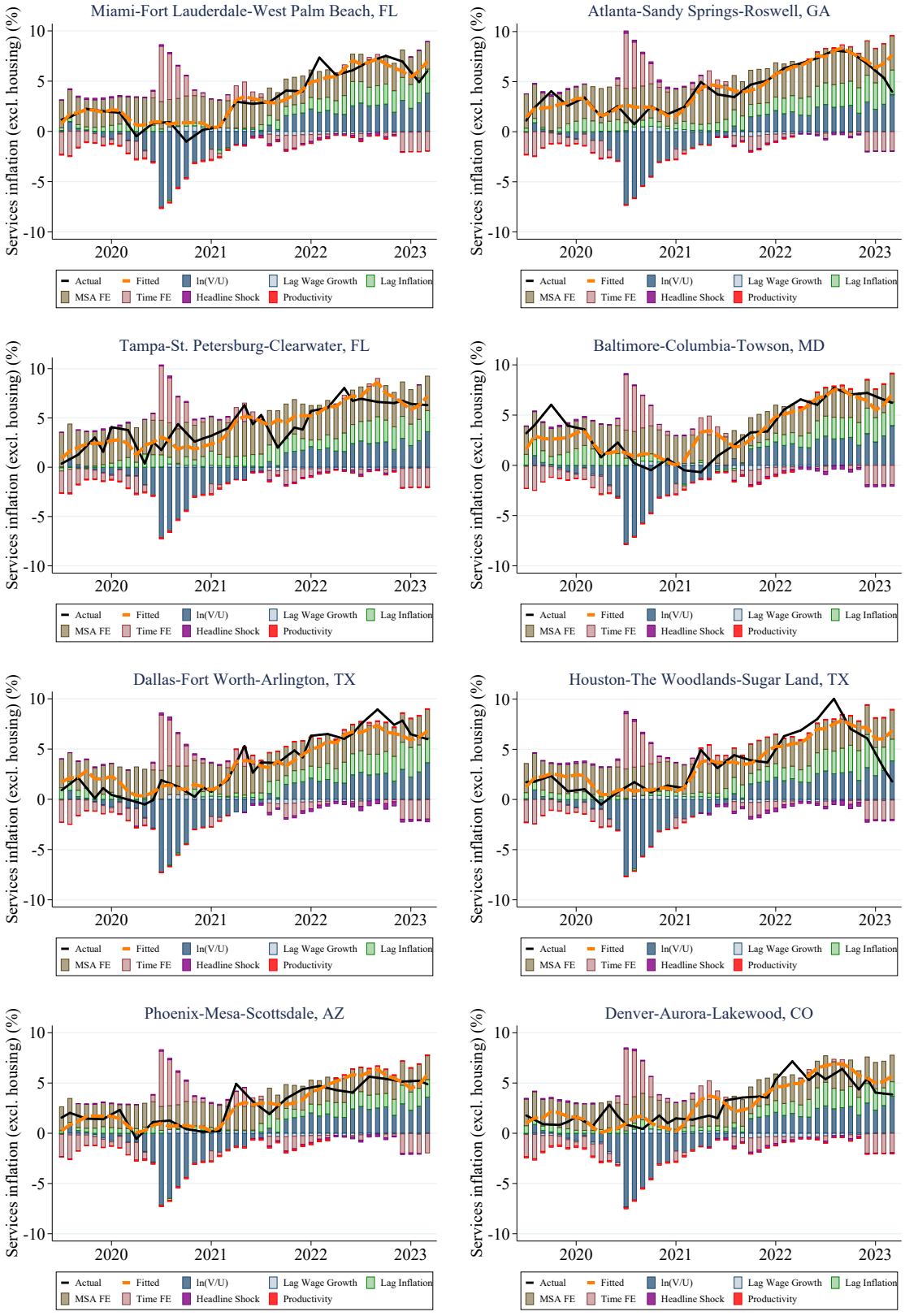


Figure 6: Predictions for services inflation (excluding housing) by MSA (Continued)

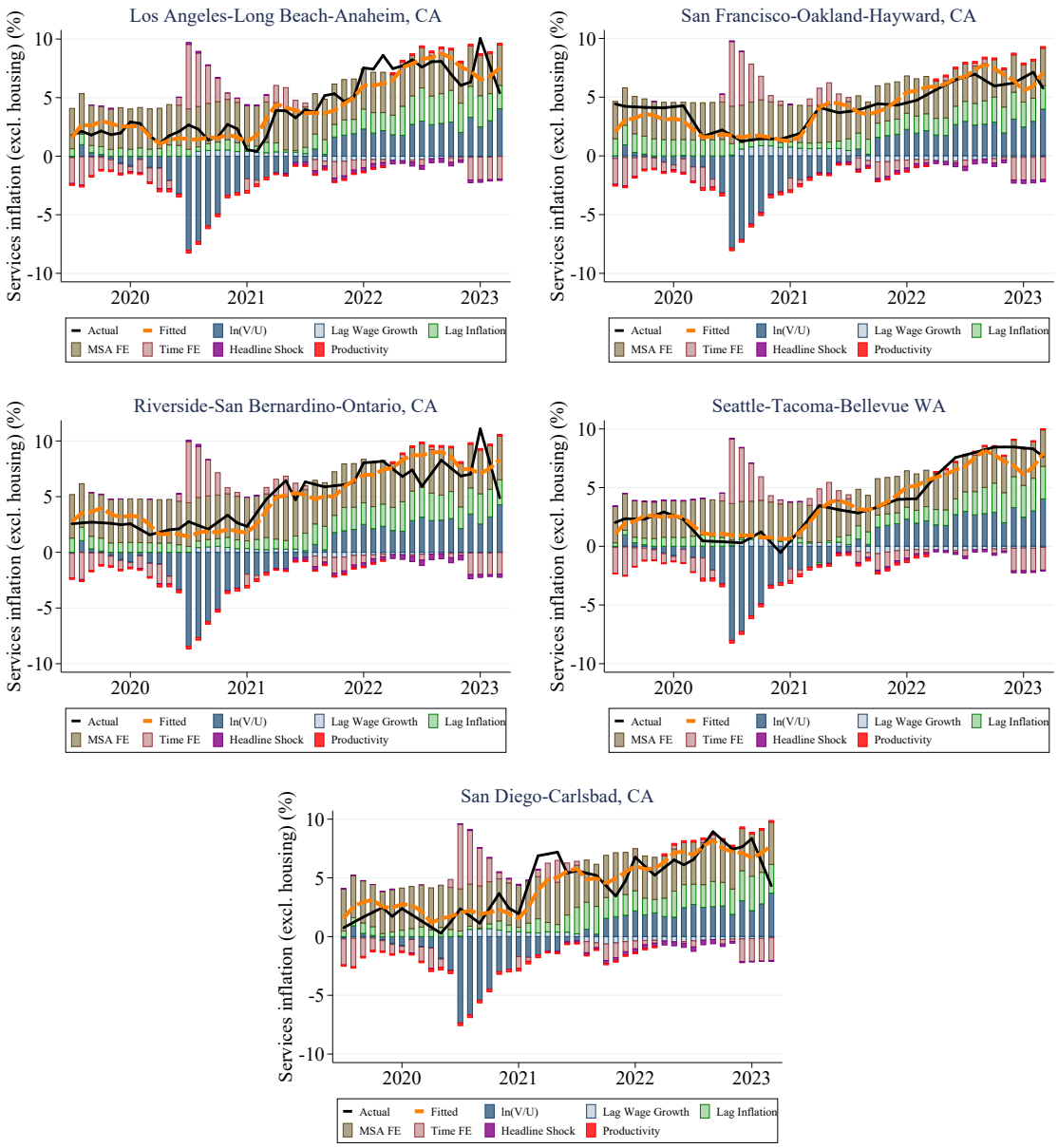


Figure 6: Predictions for services inflation (excluding housing) by MSA (Continued)

Notes: This figure plots fitted values from the LP-IV model estimated at the MSA level (Figure 4). The solid line plots actual inflation, and the dashed line plots fitted values from the LP-IV model. The bars show the contribution of each independent variable in the LP-IV model, where the contribution captures the combined effect of each variable from the first and second stages. "ln(V/U)" plots the fitted value from the Bartik shock $Shock_{i,t}$ defined in equation (2) and constructed using tradable-sector industries following the definition in Hazell et al. (2022). "Lag Wage Growth" contains fitted values from $w_{i,t-k}$, "Lag Inflation" from $y_{i,t-k}$, "MSA FE" from $\eta^h_{1,i}$ and $\eta^h_{2,i}$, "Time FE" from $\alpha^h_{1,t}$ and $\alpha^h_{2,t}$, and "Headline Shock" and "Productivity" from the controls in $X_{i,s,t}$. Source: BLS, Homebase, and authors' calculations.

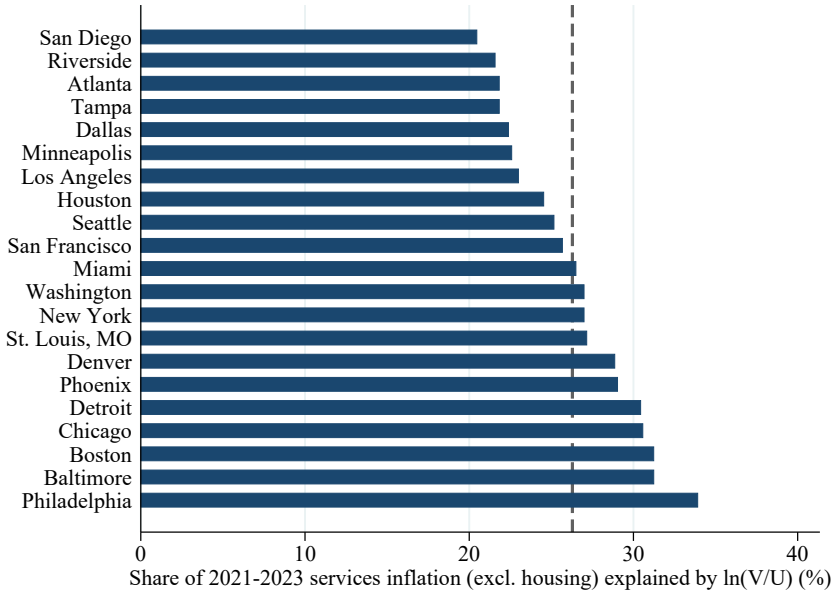


Figure 7: Share of services inflation (excluding housing) explained by ln(V/U)

Notes: This figure plots the average share of inflation explained by ln(V/U) based on fitted values of the LP-IV model estimated at the MSA level at the 3-month horizon (Figure 4):

$$y_{i,t}^{h} = \alpha_{2,t}^{h} + \eta_{2,i}^{h} + \beta_{2}^{h}\widehat{w}_{i,t} + \sum_{k=1}^{K}\gamma_{2,k}^{h}w_{i,t-k} + \sum_{k=1}^{K}\delta_{2,k}^{h}y_{i,t-k} + \gamma_{2}^{h}X_{i,s,t} + u_{2,i,t}^{h}$$

$$w_{i,t} = \alpha_{1,t}^{h} + \eta_{1,i}^{h} + \beta_{1}^{h}Shock_{i,t} + \sum_{k=1}^{K}\gamma_{1,k}^{h}w_{i,t-k} + \sum_{k=1}^{K}\delta_{1,k}^{h}y_{i,t-k} + \gamma_{1}^{h}X_{i,s,t} + u_{1,i,t}^{h}$$

where i, s, and t denote MSA, state, and month respectively. $h = 3$ denotes the estimation horizon. Share of services inflation (excluding housing) explained by ln(V/U) is calculated at the MSA level as the average fitted value for ln(V/U) over January 2021 to March 2023, as a share of the average actual services inflation (excluding housing) over the same period. Black dashed line shows the average across MSAs. The fitted value for ln(V/U) refers to the Bartik shock $Shock_{i,t}$ defined in equation (2) and constructed using tradable-sector industries following the definition in Hazell et al. (2022). Source: BLS, Homebase, and authors' calculations.

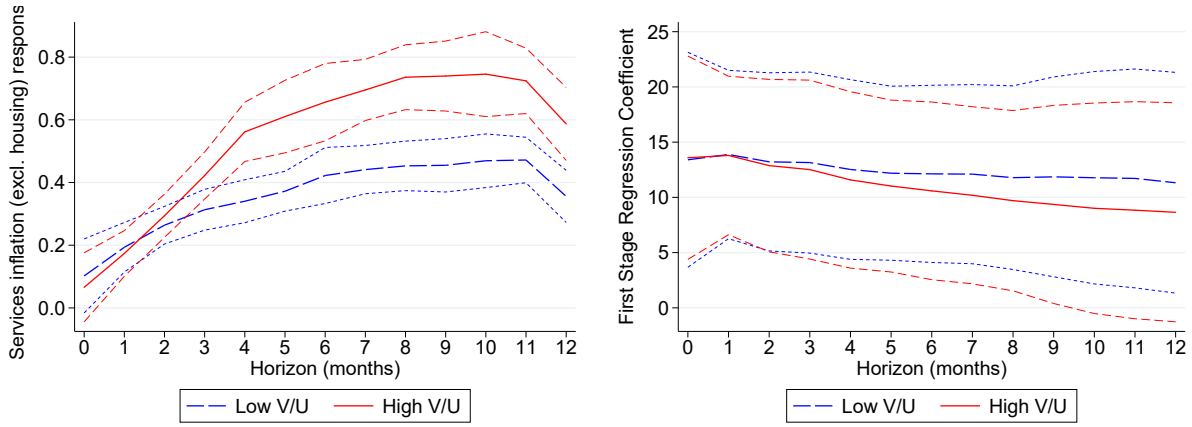


Figure 8: Services inflation (excluding housing) response, labor shortage subsample

Notes: This figure plots first and second stage estimates from the LP-IV model in equation (1) estimated over subsamples split by whether the MSA is experiencing a labor shortage ("High V/U") or otherwise ("Low V/U"):

$$y_{i,t}^h = \alpha_t^h + \eta_i^h + \beta^h w_{i,t} + \sum_{k=1}^{K} \gamma_k^h w_{i,t-k} + \sum_{k=1}^{K} \delta_k^h y_{i,t-k} + \gamma^h X_{i,s,t} + u_{i,t}^h$$

where i, s, and t denote MSA, state, and month respectively. A labor shortage refers to a situation where the vacancy-to-unemployment ratio $V_{i,t}/U_{i,t}$ is greater than 1. $h = 0, 1, \ldots, H$ denotes the estimation horizon. We control for lags up to $K = 3$. $y_{i,t}$ is year-on-year inflation of MSA i in month t measured by the logarithmic difference in the prices for services excluding housing, $ln(prices)_{i,t} - ln(prices)_{i,t-12}$. The dependent variable $y_{i,t}^h$ denotes year-on-year inflation at $t + h$. α_t^h is a time fixed effect. η_i^h is an MSA fixed effect. $w_{i,t}$ is our independent variable of interest, year-on-year wage growth in the service sector from Homebase measured by logarithmic difference, $ln(wage)_{i,t} - ln(wage)_{i,t-12}$. We instrument wage growth using the Bartik shock defined in equation (2) and constructed using tradable-sector industries following the definition in Hazell et al. (2022). We also include a vector of controls $X_{i,s,t}$ at their native geographic and temporal levels, including labor productivity (from BLS) to proxy for local labor demand, headline inflation shocks defined as the difference between year-on-year headline and core excluding food and energy inflation, and residuals from the regressions for previous horizons. The sample is a monthly panel of MSAs over January 2019 to December 2022. Observations are weighted by the average size of the labor force in each MSA in 2019. We cluster the standard errors $u_{i,t}^h$ at the MSA level. The solid line shows the point estimates and the dashed lines show the 90 percent confidence interval. Statistical significance is accessed using weak-IV-robust (Anderson–Rubin) p-values. The full-sample pooled-interacted Kleibergen-Paap rk Wald F is 3.60 at $h = 3$ and 2.74 at $h = 10$. Source: BLS, Homebase, and authors' calculations.

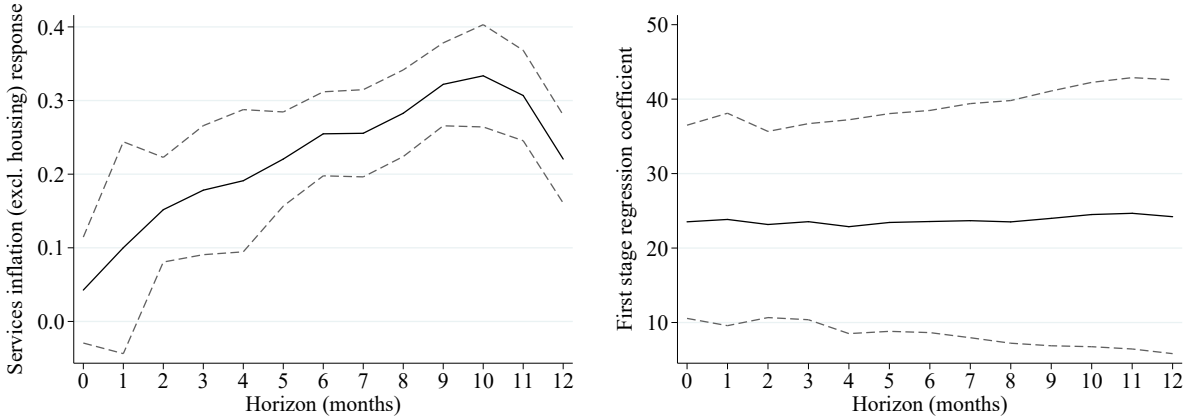


Figure 9: Services inflation (excluding housing) response, leave-one-out (LOO) estimator

Notes: This figure plots first and second stage estimates from the LP-IV model in equation (1):

$$y_{i,t}^h = \alpha_t^h + \eta_i^h + \beta^h w_{i,t} + \sum_{k=1}^{K} \gamma_k^h w_{i,t-k} + \sum_{k=1}^{K} \delta_k^h y_{i,t-k} + \gamma^h X_{i,s,t} + u_{i,t}^h$$

where i, s, and t denote MSA, state, and month respectively. $h = 0, 1, \ldots, H$ denotes the estimation horizon. We control for lags up to $K = 3$. $y_{i,t}$ is year-on-year inflation of MSA i in month t measured by the logarithmic difference in the prices for services excluding housing, $ln(prices)_{i,t} - ln(prices)_{i,t-12}$. The dependent variable $y_{i,t}^h$ denotes year-on-year inflation at $t + h$. α_t^h is a time fixed effect. η_i^h is an MSA fixed effect. $w_{i,t}$ is our independent variable of interest, year-on-year wage growth in the service sector from Homebase measured by logarithmic difference, $ln(wage)_{i,t} - ln(wage)_{i,t-12}$. We instrument wage growth using the Bartik shock defined in equation (2) and constructed using a leave-one-out (LOO) approach following Borusyak, Hull and Jaravel (2022). For each MSA, the national industry shock is computed excluding that MSA's own contribution. We also include a vector of controls $X_{i,s,t}$ at their native geographic and temporal levels, including labor productivity (from BLS) to proxy for local labor demand, headline inflation shocks defined as the difference between year-on-year headline and core excluding food and energy inflation, and residuals from the regressions for previous horizons. The sample is a monthly panel of MSAs over January 2019 to December 2022. Observations are weighted by the average size of the labor force in each MSA in 2019. We cluster the standard errors $u_{i,t}^h$ at the MSA level. The solid line shows the point estimates and the dashed lines show the 90 percent confidence interval. Source: BLS, Homebase, and authors' calculations.

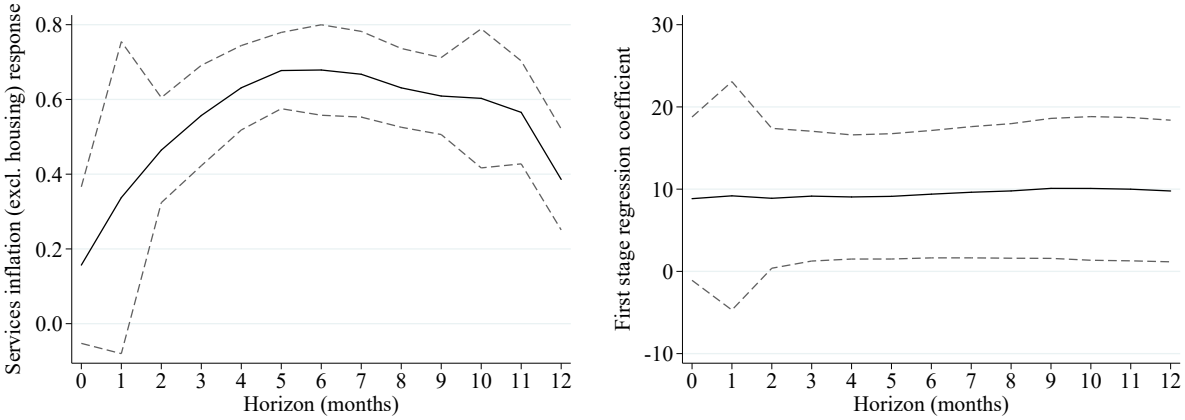


Figure 10: Services inflation (excluding housing) response, MSA level model with region-time fixed effects

Notes: This figure plots first and second stage estimates from the following LP-IV model:

$$y_{i,t}^{h} = \alpha_{r,t}^{h} + \eta_{i}^{h} + \beta^{h} w_{i,t} + \sum_{k=1}^{K} \gamma_{k}^{h} w_{i,t-k} + \sum_{k=1}^{K} \delta_{k}^{h} y_{i,t-k} + \gamma^{h} X_{i,s,t} + u_{i,t}^{h}$$

where i, s, r, and t denote MSA, state, region, and month respectively. $h = 0, 1, \ldots, H$ denotes the estimation horizon. We control for lags up to $K = 3$. $y_{i,t}$ is year-on-year inflation of MSA i in month t measured by the logarithmic difference in the prices for services excluding housing, $ln(prices)_{i,t} - ln(prices)_{i,t-12}$. The dependent variable $y_{i,t}^{h}$ denotes year-on-year inflation at $t+h$. $\alpha_{r,t}^{h}$ is a region-time fixed effect. η_{i}^{h} is an MSA fixed effect. $w_{i,t}$ is our independent variable of interest, year-on-year wage growth in the service sector from Homebase measured by logarithmic difference, $ln(wage)_{i,t} - ln(wage)_{i,t-12}$. We instrument wage growth using the Bartik shock defined in equation (2) and constructed using tradable-sector industries following the definition in Hazell et al. (2022). We also include a vector of controls $X_{i,s,t}$ at their native geographic and temporal levels, including labor productivity (from BLS) to proxy for local labor demand, headline inflation shocks defined as the difference between year-on-year headline and core excluding food and energy inflation, and residuals from the regressions for previous horizons. The sample is a monthly panel of MSAs over January 2019 to December 2022. MSAs are mapped to states based on the principal state to which the MSA belongs. States are categorized into four Census Bureau designated regions. Observations are weighted by the average size of the labor force in each MSA in 2019. We cluster the standard errors $u_{i,t}^{h}$ at the MSA level. The solid line shows the point estimates and the dashed lines show the 90 percent confidence interval. Source: BLS, Homebase, and authors' calculations.

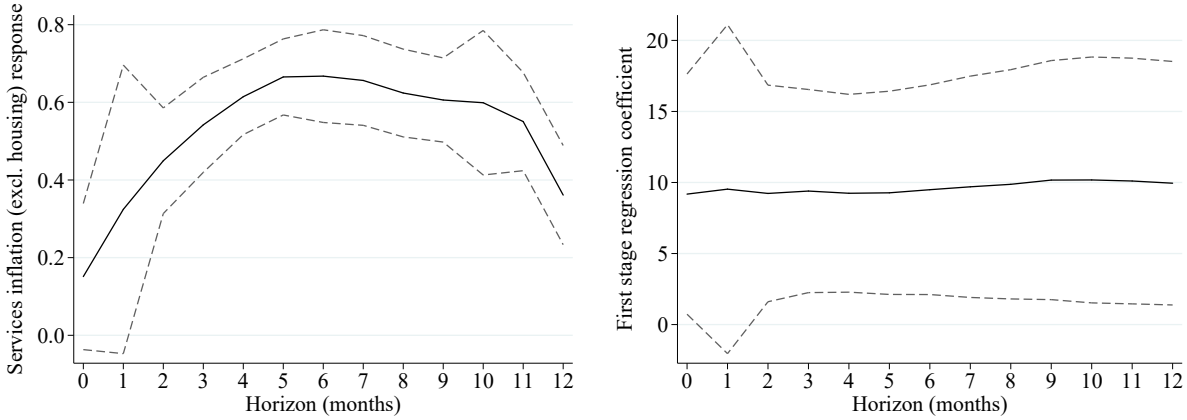


Figure 11: Services inflation (excluding housing) response, MSA level model with region-time fixed effects, household income, and housing wealth

Notes: This figure plots first and second stage estimates from the following LP-IV model:

$$y_{i,t}^{h} = \alpha_{r,t}^{h} + \eta_{i}^{h} + \beta^{h} w_{i,t} + \sum_{k=1}^{K} \gamma_{k}^{h} w_{i,t-k} + \sum_{k=1}^{K} \delta_{k}^{h} y_{i,t-k} + \gamma^{h} X_{i,s,t} + u_{i,t}^{h}$$

where i, s, r, and t denote MSA, state, region, and month respectively. $h = 0, 1, \ldots, H$ denotes the estimation horizon. We control for lags up to $K = 3$. $y_{i,t}$ is year-on-year inflation of MSA i in month t measured by the logarithmic difference in the prices for services excluding housing, $ln(prices)_{i,t} - ln(prices)_{i,t-12}$. The dependent variable $y_{i,t}^{h}$ denotes year-on-year inflation at $t+h$. $\alpha_{r,t}^{h}$ is a region-time fixed effect. η_{i}^{h} is an MSA fixed effect. $w_{i,t}$ is our independent variable of interest, year-on-year wage growth in the service sector from Homebase measured by logarithmic difference, $ln(wage)_{i,t} - ln(wage)_{i,t-12}$. We instrument wage growth using the Bartik shock defined in equation (2) and constructed using tradable-sector industries following the definition in Hazell et al. (2022). We also include a vector of controls $X_{i,s,t}$ at their native geographic and temporal levels, including labor productivity (from BLS) to proxy for local labor demand, headline inflation shocks defined as the difference between year-on-year headline and core excluding food and energy inflation, log household income per capita, log housing wealth defined as the product between Census homeownership rates and Freddie Mac House Price Index, and residuals from the regressions for previous horizons. The sample is a monthly panel of MSAs over January 2019 to December 2022. MSAs are mapped to states based on the principal state to which the MSA belongs. States are categorized into four Census Bureau designated regions. Observations are weighted by the average size of the labor force in each MSA in 2019. We cluster the standard errors $u_{i,t}^{h}$ at the MSA level. The solid line shows the point estimates and the dashed lines show the 90 percent confidence interval. Source: BLS, Homebase, and authors' calculations.

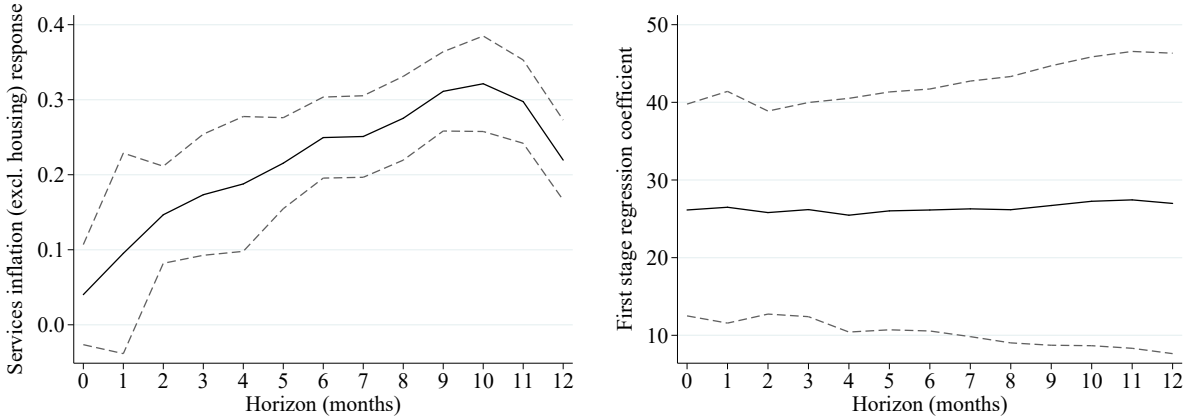


Figure 12: Services inflation (excluding housing) response, Bartik instrument using all sectors

Notes: This figure plots first and second stage estimates from the LP-IV model in equation (1):

$$y_{i,t}^h = \alpha_t^h + \eta_i^h + \beta^h w_{i,t} + \sum_{k=1}^{K} \gamma_k^h w_{i,t-k} + \sum_{k=1}^{K} \delta_k^h y_{i,t-k} + \gamma^h X_{i,s,t} + u_{i,t}^h$$

where i, s, and t denote MSA, state, and month respectively. $h = 0, 1, \ldots, H$ denotes the estimation horizon. We control for lags up to $K = 3$. $y_{i,t}$ is year-on-year inflation of MSA i in month t measured by the logarithmic difference in the prices for services excluding housing, $ln(prices)_{i,t} - ln(prices)_{i,t-12}$. The dependent variable $y_{i,t}^h$ denotes year-on-year inflation at $t + h$. α_t^h is a time fixed effect. η_i^h is an MSA fixed effect. $w_{i,t}$ is our independent variable of interest, year-on-year wage growth in the service sector from Homebase measured by logarithmic difference, $ln(wage)_{i,t} - ln(wage)_{i,t-12}$. We instrument wage growth using the Bartik shock for the log vacancy-to-unemployment ratio, which is defined in equation (2), constructed using all 3-digit NAICS industries. Because this instrument includes non-tradable service-sector shifts, the baseline tradable-exclusion rationale no longer applies, so this figure is a descriptive sensitivity check rather than a causal estimate. We also include a vector of controls $X_{i,s,t}$ at their native geographic and temporal levels, including labor productivity (from BLS) to proxy for local labor demand, headline inflation shocks defined as the difference between year-on-year headline and core excluding food and energy inflation, and residuals from the regressions for previous horizons. The sample is a monthly panel of MSAs over January 2019 to December 2022. Observations are weighted by the average size of the labor force in each MSA in 2019. We cluster the standard errors $u_{i,t}^h$ at the MSA level. The solid line shows the point estimates and the dashed lines show the 90 percent confidence interval. Source: BLS, Homebase, and authors' calculations.

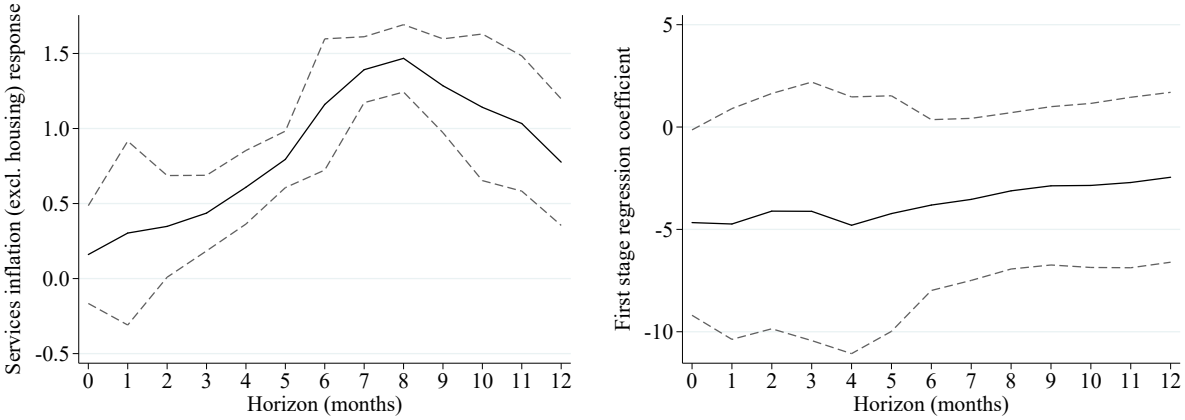


Figure 13: Services inflation (excluding housing) response, Bartik shock based on unemployment rate

Notes: This figure plots first and second stage estimates from the LP-IV model in equation (1):

$$y_{i,t}^h = \alpha_t^h + \eta_i^h + \beta^h w_{i,t} + \sum_{k=1}^{K} \gamma_k^h w_{i,t-k} + \sum_{k=1}^{K} \delta_k^h y_{i,t-k} + \gamma^h X_{i,s,t} + u_{i,t}^h$$

where i, s, and t denote MSA, state, and month respectively. $h = 0, 1, \ldots, H$ denotes the estimation horizon. We control for lags up to $K = 3$. $y_{i,t}$ is year-on-year inflation of MSA i in month t measured by the logarithmic difference in the prices for services excluding housing, $ln(prices)_{i,t} - ln(prices)_{i,t-12}$. The dependent variable $y_{i,t}^h$ denotes year-on-year inflation at $t + h$. α_t^h is a time fixed effect. η_i^h is an MSA fixed effect. $w_{i,t}$ is our independent variable of interest, year-on-year wage growth in the service sector from Homebase measured by logarithmic difference, $ln(wage)_{i,t} - ln(wage)_{i,t-12}$. We instrument wage growth using a Bartik shock constructed using the log of unemployment rate in tradable-sector industries following the definition in Hazell et al. (2022). We also include a vector of controls $X_{i,s,t}$ at their native geographic and temporal levels, including labor productivity (from BLS) to proxy for local labor demand, headline inflation shocks defined as the difference between year-on-year headline and core excluding food and energy inflation, and residuals from the regressions for previous horizons. The sample is a monthly panel of MSAs over January 2019 to December 2022. Observations are weighted by the average size of the labor force in each MSA in 2019. We cluster the standard errors $u_{i,t}^h$ at the MSA level. The solid line shows the point estimates and the dashed lines show the 90 percent confidence interval.Statistical significance is accessed using weak-IV-robust (Anderson–Rubin) p-values. The first-stage Kleibergen-Paap rk Wald F statistic at the three-month horizon is 1.35. Source: BLS, Homebase, and authors' calculations.

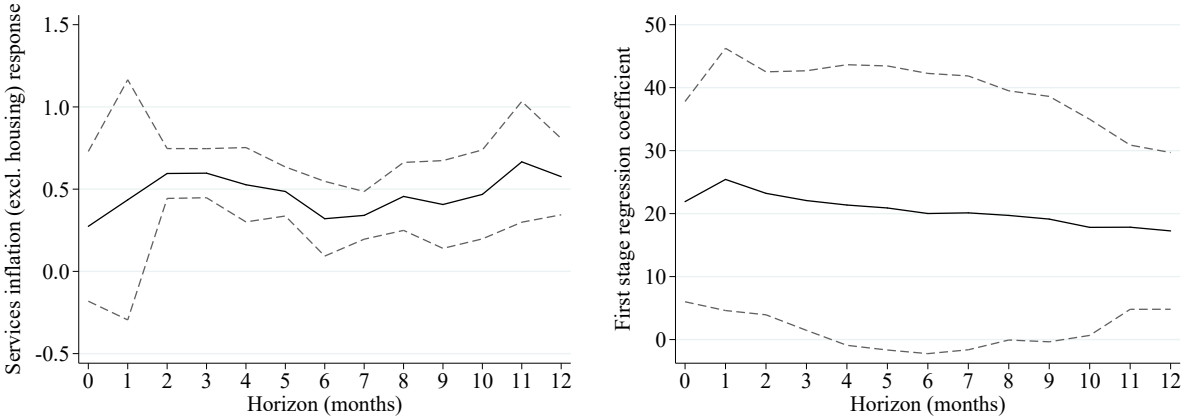


Figure 14: Services inflation (excluding housing) response, Bartik shock based on employment growth

Notes: This figure plots first and second stage estimates from the LP-IV model in equation (1):

$$y_{i,t}^h = \alpha_t^h + \eta_i^h + \beta^h w_{i,t} + \sum_{k=1}^{K} \gamma_k^h w_{i,t-k} + \sum_{k=1}^{K} \delta_k^h y_{i,t-k} + \gamma^h X_{i,s,t} + u_{i,t}^h$$

where i, s, and t denote MSA, state, and month respectively. $h = 0, 1, \ldots, H$ denotes the estimation horizon. We control for lags up to $K = 3$. $y_{i,t}$ is year-on-year inflation of MSA i in month t measured by the logarithmic difference in the prices for services excluding housing, $ln(prices)_{i,t} - ln(prices)_{i,t-12}$. The dependent variable $y_{i,t}^h$ denotes year-on-year inflation at $t + h$. α_t^h is a time fixed effect. η_i^h is an MSA fixed effect. $w_{i,t}$ is our independent variable of interest, year-on-year wage growth in the service sector from Homebase measured by logarithmic difference, $ln(wage)_{i,t} - ln(wage)_{i,t-12}$. We instrument wage growth using a Bartik shock constructed using the twelve-month log change in national tradable-industry employment, $\ln(Emp_{j,t}) - \ln(Emp_{j,t-12})$ following the definition in Hazell et al. (2022). We also include a vector of controls $X_{i,s,t}$ at their native geographic and temporal levels, including labor productivity (from BLS) to proxy for local labor demand, headline inflation shocks defined as the difference between year-on-year headline and core excluding food and energy inflation, and residuals from the regressions for previous horizons. The sample is a monthly panel of MSAs over January 2019 to December 2022. Observations are weighted by the average size of the labor force in each MSA in 2019. We cluster the standard errors $u_{i,t}^h$ at the MSA level. The solid line shows the point estimates and the dashed lines show the 90 percent confidence interval. Statistical significance is accessed using weak-IV-robust (Anderson–Rubin) p-values. The first-stage Kleibergen-Paap rk Wald F statistic at the three-month horizon is 3.46. Source: BLS, Homebase, and authors' calculations.

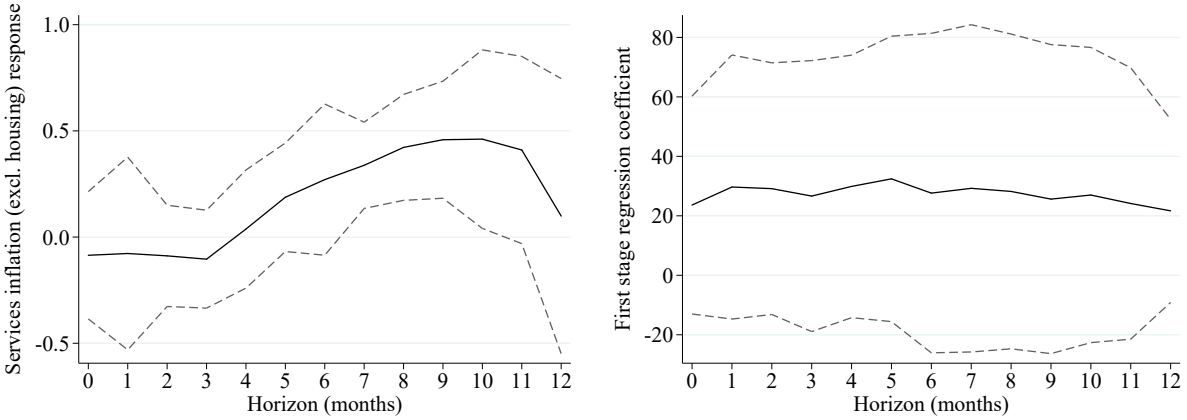


Figure 15: Services inflation (excluding housing) response, Bartik shock based on log payroll

Notes: This figure plots first and second stage estimates from the LP-IV model in equation (1):

$$y_{i,t}^h = \alpha_t^h + \eta_i^h + \beta^h w_{i,t} + \sum_{k=1}^{K} \gamma_k^h w_{i,t-k} + \sum_{k=1}^{K} \delta_k^h y_{i,t-k} + \gamma^h X_{i,s,t} + u_{i,t}^h$$

where i, s, and t denote MSA, state, and month respectively. $h = 0, 1, \ldots, H$ denotes the estimation horizon. We control for lags up to $K = 3$. $y_{i,t}$ is year-on-year inflation of MSA i in month t measured by the logarithmic difference in the prices for services excluding housing, $ln(prices)_{i,t} - ln(prices)_{i,t-12}$. The dependent variable $y_{i,t}^h$ denotes year-on-year inflation at $t + h$. α_t^h is a time fixed effect. η_i^h is an MSA fixed effect. $w_{i,t}$ is our independent variable of interest, year-on-year wage growth in the service sector from Homebase measured by logarithmic difference, $ln(wage)_{i,t} - ln(wage)_{i,t-12}$. We instrument local labor market wage growth using a Bartik shock constructed using the log level of national tradable-industry payroll, $\ln(Payroll_{j,t})$ following the definition in Hazell et al. (2022). We also include a vector of controls $X_{i,s,t}$ at their native geographic and temporal levels, including labor productivity (from BLS) to proxy for local labor demand, headline inflation shocks defined as the difference between year-on-year headline and core excluding food and energy inflation, and residuals from the regressions for previous horizons. The sample is a monthly panel of MSAs over January 2019 to December 2022. Observations are weighted by the average size of the labor force in each MSA in 2019. We cluster the standard errors $u_{i,t}^h$ at the MSA level. The solid line shows the point estimates and the dashed lines show the 90 percent confidence interval. Source: BLS, Homebase, and authors' calculations. The first-stage Kleibergen-Paap rk Wald F statistic at the three-month horizon is 0.90.

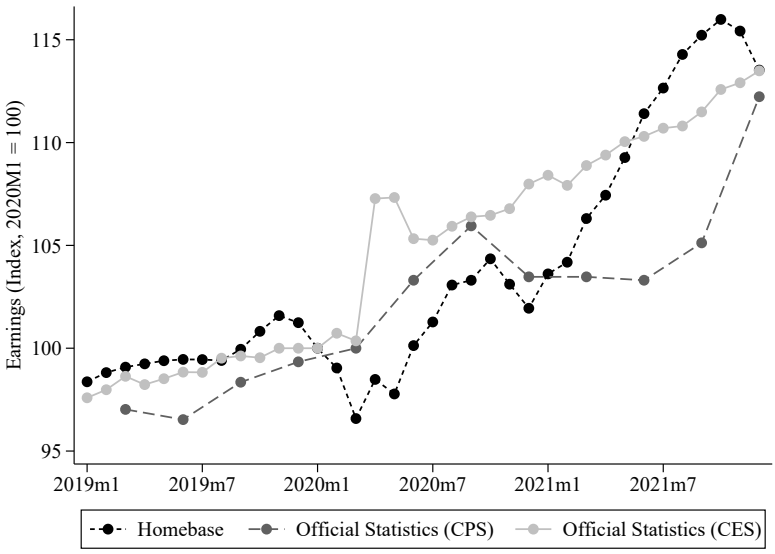


Figure 16: Average wage trend: a comparison

Notes: This figure plots national average earnings in the Homebase sample against official statistics. We consider two sources of official statistics, both from the BLS. The Current Population Survey (CPS) line plots the median weekly nominal earnings of wage and salary workers in service occupations of 16 years and over (LEU0254543400Q). The Current Employment Statistics (CES) line plots the average weekly earnings of all employees from the private service-providing sector (CES0800000011). The Homebase sample excludes workers that were on temporary layoffs, defined as workers that were initially employed (reported positive earnings), were laid off during the middle of the sample (reported zero earnings), and later returned to work again before the end of the sample. The sample includes all workers that were in the database for at least one month between 2019 and 2021. These figures therefore validate wage and earnings dynamics in a related Homebase sample rather than the exact wage variable used in the LP-IV, which averages employed workers through 2022. Each series is expressed as an index relative to its value in January 2020 (and Q1 2020 for CPS). The correlation between Homebase and CPS series is 0.73; the correlation between Homebase and CES series is 0.87. Source: BLS, Homebase, and authors' calculations.

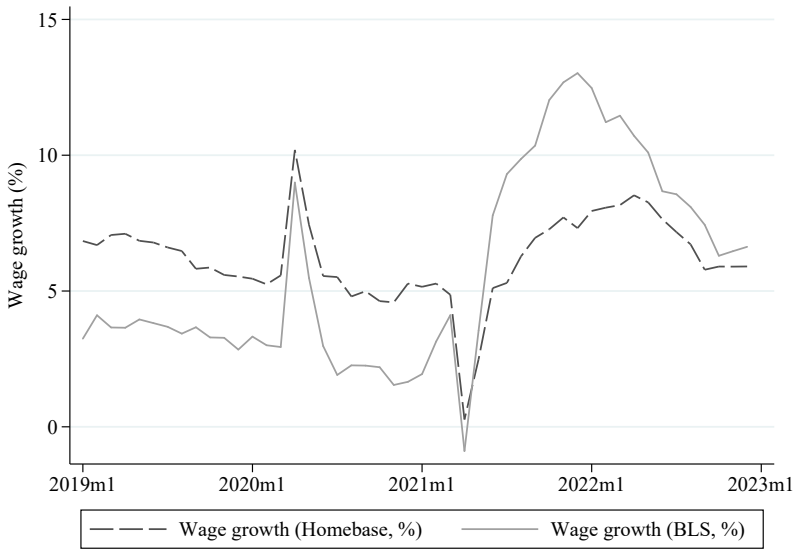


Figure 17: Wage growth: a comparison

Notes: This figure plots year-on-year average hourly wage growth from Homebase and the average hourly earnings for the leisure and hospitality industry (CES7000000003) from the Current Employment Statistics (CES). The Homebase and CES series have a correlation of 0.68. Source: BLS, Homebase, and authors' calculations.

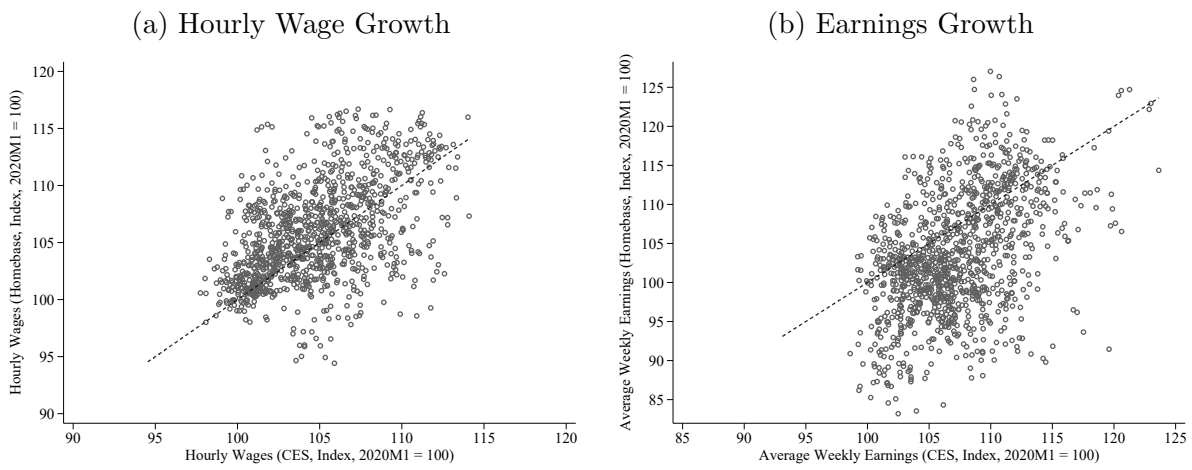


Figure 18: State-level wage and earnings growth: a comparison

Notes: This figure compares growth in hourly wages and total earnings relative to January 2020 between the Homebase data and official statistics (CES), by month and state. The Current Employment Statistics (CES) data use state-level average weekly earnings for earnings (SMU[2-digit state FIPS code]000000800000011), average hourly earnings for hourly wages (SMU[2-digit state FIPS code]000000800000003), and number of employees for employment (SMS[2-digit state FIPS code]000000800000001), for all employees from the private service-providing sector. The correlation between Homebase and CES is 0.51 for hourly wages and 0.45 for earnings. These correlations are computed across state-month observations expressed as indices relative to January 2020, so they reflect comovement in indexed levels rather than in month-to-month changes. Dashed line plots the 45 degree line from the origin. The Homebase sample excludes workers that were on temporary layoffs, defined as workers that were initially employed (reported positive earnings), were laid off during the middle of the sample (reported zero earnings), and later returned to work again before the end of the sample. The sample includes all workers that were in the database for at least one month between 2019 and 2021. Source: BLS, Homebase, and authors' calculations.

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A Appendix

A.1 Cross-sectional evidence from official statistics

We estimate our LP-IV model using wage data from the QCEW at the MSA quarterly level:

$$y_{i,t}^h = \alpha_t^h + \eta_i^h + \beta^h w_{i,t} + \sum_{k=1}^{K} \gamma_k^h w_{i,t-k} + \sum_{k=1}^{K} \delta_k^h y_{i,t-k} + \gamma^h X_{i,s,t} + u_{i,t}^h, \tag{4}$$

where i, s, and t denote MSA, state, and quarter, respectively. $h = 0, 1, \ldots, H$ denotes the estimation horizon. We control for lags up to $K = 1$. $y_{i,t}$ is year-on-year inflation of MSA i in quarter t measured by the logarithmic difference in the prices for services excluding housing, $ln(prices)_{i,t} - ln(prices)_{i,t-4}$. The dependent variable $y_{i,t}^h$ denotes year-on-year inflation h quarters ahead. α_t^h is a time fixed effect. η_i^h is an MSA fixed effect. $w_{i,t}$ is our independent variable of interest, year-on-year log growth of total quarterly payroll in the leisure and hospitality sector from the QCEW, $ln(payroll)_{i,t} - ln(payroll)_{i,t-4}$. We instrument payroll growth using the Bartik shock defined in equation (2). The monthly shock is converted to quarterly frequencies by taking the average within each calendar quarter. We also include a vector of controls $X_{i,s,t}$ at the corresponding geographic and temporal levels, including labor productivity (from BLS) to proxy for local labor demand, headline inflation shocks defined as the difference between year-on-year headline and core excluding food and energy inflation, and residuals from the regressions for previous horizons. Observations are weighted by the average size of the labor force in each MSA in 2019. We cluster the standard errors $u_{i,t}^h$ at the MSA level. The sample period is from 2019Q1 to 2022Q4.[21] Figure 19 plots the estimated coefficients and Figure 20 decomposes estimation results at the 1-quarter horizon.

A.2 Time-series evidence from official statistics

We use official statistics from BLS on vacancy, unemployment, and prices at the national level to estimate a dynamic Phillips curve using the following LP specification:

$$y_t^h = \alpha^h + \beta^h \ln \theta_t + \sum_{k=1}^{K} \gamma_{\theta,k}^h \ln \theta_{t-k} + \sum_{k=1}^{K} \gamma_{y,k}^h y_{t-k} + \gamma_x^h X_t + u_t^h,$$

for monthly time t and horizon $h = 0, 1, \ldots, 36$ months. y_t is year-on-year services inflation excluding housing in month t, computed as logarithmic changes. The dependent variable y_t^h denotes inflation h months ahead of time t. $\ln \theta_t$ is our independent variable of interest: log V/U from Barnichon (2010) and updated from JOLTS. X_t is a vector of control variables including labor productivity (from BLS), headline inflation shocks, and inflation expectations. Headline inflation shock is defined as the difference between year-on-year headline and core excluding food and energy inflation. Inflation expectations are measured using the 2-year inflation expectations from the Federal Reserve Bank of Cleveland, combined with 12-month inflation expectations from the Livingston survey before 1982. We include $K = 3$ lags of inflation and log V/U, with other controls (labor productivity, the headline inflation shock, and inflation expectations) entering contemporaneously. The sample period is January 1986 to December 2023. Figure 21 plots the estimated coefficients.

We explore the time variation in β^h by keeping the horizon fixed at $h = 12$ months when the coefficient peaks and running rolling regressions with a 3-year window. Figure 22 plots the estimated coefficients, which shows substantial time-variation. The estimate for the 2021-2023 period is about 4.0, which is very similar to the estimates in Benigno and Eggertsson (2023) for high V/U.

To illustrate the important role of labor market tightness for services excluding housing inflation, Figure 23 plots the fitted values (the "Fitted" line) from the rolling regression above as well as counterfactual fitted values (the "Fitted w/o ln(V/U)" line) from the same regression without a tight labor market by imposing the contemporaneous and 1-3 lagged values of V/U to 1, while retaining other controls. The decomposition is in-sample, and the contribution of labor market tightness is the gap between the "Fitted" and "Fitted

[21] Wage data start from Q1 1990 but the Bartik shock data only starts in Q1 2019.

w/o ln(V/U)" lines. The "Fitted" line offers better fit of actual inflation than the "Fitted w/o ln(V/U)" line, consistent with an important role of labor market tightness in explaining the post-pandemic inflation surge. The counterfactual fitted path peaks in early 2021 and falls below 2 percent in early 2023.

A.3 Constructing nationally representative vacancies

Data on vacancies from Indeed may not necessarily be nationally representative because some job openings may not be posted online. We benchmark the three-digit Indeed series to JOLTS using vacancy-to-unemployment ratios. For each three-digit industry, we regress the corresponding two-digit industry log V/U ratio using JOLTS vacancies and CPS unemployment on the three-digit log V/U ratio using Indeed vacancies and CPS unemployment. We use the fitted log V/U values, add back three-digit log unemployment, and exponentiate to obtain the benchmarked vacancy series. This adjustment thus thus benchmarks vacancies from Indeed to the corresponding industry in JOLTS.

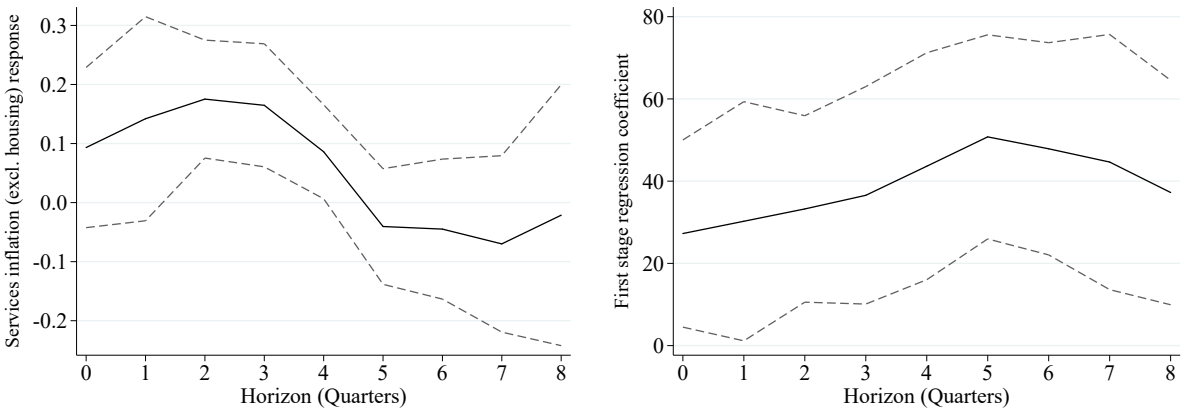


Figure 19: Services inflation (excluding housing) response, MSA level using QCEW payroll

Notes: This figure plots first and second stage estimates from the following LP-IV model:

$$y_{i,t}^{h} = \alpha_t^h + \eta_i^h + \beta^h w_{i,t} + \sum_{k=1}^{K} \gamma_k^h w_{i,t-k} + \sum_{k=1}^{K} \delta_k^h y_{i,t-k} + \gamma^h X_{i,s,t} + u_{i,t}^h$$

where i, s, and t denote MSA, state, and quarter, respectively. $h = 0, 1, \ldots, H$ denotes the estimation horizon. We control for lags up to $K = 1$. $y_{i,t}$ is year-on-year inflation of MSA i in quarter t measured by the logarithmic difference in the prices for services excluding housing, $ln(prices)_{i,t} - ln(prices)_{i,t-4}$. The dependent variable $y_{i,t}^h$ denotes year-on-year inflation h quarters ahead. α_t^h is a time fixed effect. η_i^h is an MSA fixed effect. $w_{i,t}$ is our independent variable of interest, year-on-year log growth of total quarterly leisure-and-hospitality payroll from the QCEW, $ln(payroll)_{i,t} - ln(payroll)_{i,t-4}$, without an employment or hours denominator. We instrument payroll growth using the Bartik shock for the log vacancy-to-unemployment ratio, which is defined in equation (2), constructed using only tradable-sector industries following the definition in Hazell et al. (2022). We also include a vector of controls $X_{i,s,t}$ at their native geographic and temporal levels, including labor productivity (from BLS) to proxy for local labor demand, headline inflation shocks defined as the difference between year-on-year headline and core excluding food and energy inflation, and residuals from the regressions for previous horizons. The sample is a quarterly panel of MSAs over Q1 2019 to Q4 2022. Observations are weighted by the average size of the labor force in each MSA in 2019. We cluster the standard errors $u_{i,t}^h$ at the MSA level. The solid line shows the point estimates and the dashed lines show the 90 percent confidence interval. Source: BLS, QCEW, and authors' calculations. The first-stage coefficient concerns contemporaneous growth for each horizon-specific estimation sample and controls, not growth accumulated over the horizon.

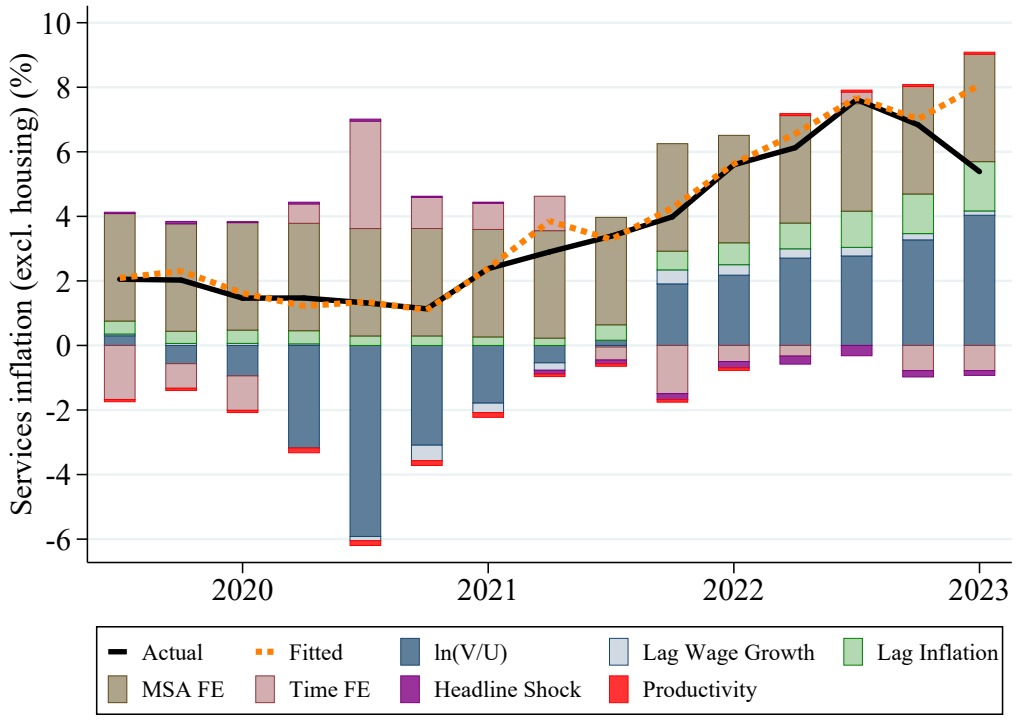


Figure 20: Predictions for services inflation (excluding housing), MSA level using QCEW payroll

Notes: This figure plots fitted values from the LP-IV model estimated at the MSA level (Figure 19):

$$y_{i,t}^h = \alpha_{2,t}^h + \eta_{2,i}^h + \beta_2^h \widehat{w}_{i,t} + \sum_{k=1}^{K} \gamma_{2,k}^h w_{i,t-k} + \sum_{k=1}^{K} \delta_{2,k}^h y_{i,t-k} + \gamma_2^h X_{i,s,t} + u_{2,i,t}^h$$

$$w_{i,t} = \alpha_{1,t}^h + \eta_{1,i}^h + \beta_1^h Shock_{i,t} + \sum_{k=1}^{K} \gamma_{1,k}^h w_{i,t-k} + \sum_{k=1}^{K} \delta_{1,k}^h y_{i,t-k} + \gamma_1^h X_{i,s,t} + u_{1,i,t}^h$$

where i, s, and t denote MSA, state, and quarter, respectively. $h = 1$ denotes the estimation horizon. The solid line plots actual inflation, and the dashed line plots fitted values from the LP-IV model. All values are unweighted averages across MSAs in the sample. The bars show the contribution of each independent variable in the LP-IV model, where the contribution captures the combined effect of each variable from the first and second stages. "ln(V/U)" plots the fitted value from the Bartik shock $Shock_{i,t}$ for the log vacancy-to-unemployment ratio defined in equation (2), constructed using only tradable-sector industries following the definition in Hazell et al. (2022). "Lag Wage Growth" contains fitted values from $w_{i,t-k}$, "Lag Inflation" from $y_{i,t-k}$, "MSA FE" from $\eta_{1,i}^h$ and $\eta_{2,i}^h$, "Time FE" from $\alpha_{1,t}^h$ and $\alpha_{2,t}^h$, and "Headline Shock" and "Productivity" from the controls in $X_{i,s,t}$. Source: BLS, QCEW, and authors' calculations.

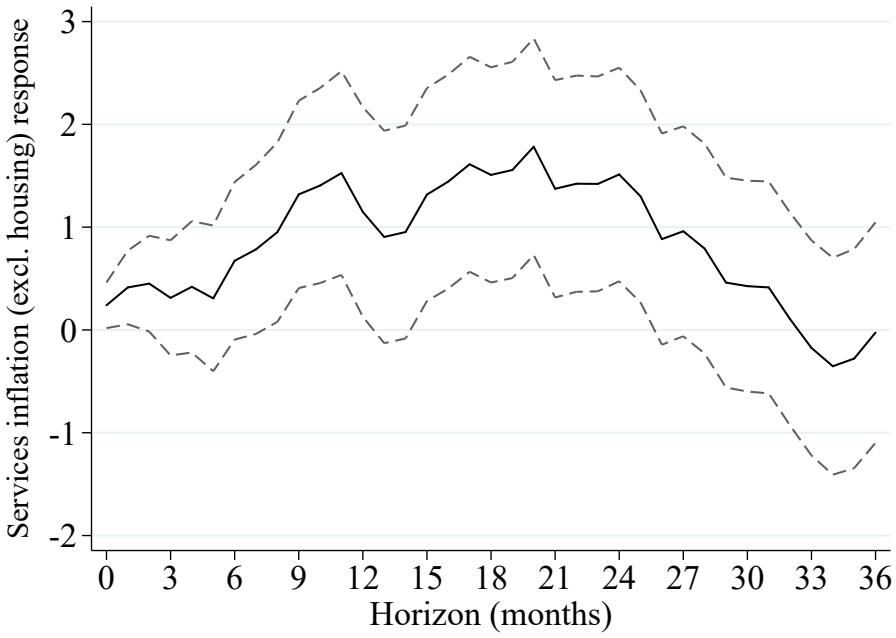


Figure 21: Services inflation (excluding housing) response, national level official statistics

Notes: This figure plots estimates of β^h from the following local projection (LP) model:

$$y_t^h = \alpha^h + \beta^h \ln \theta_t + \sum_{k=1}^{K} \gamma_{\theta,k}^h \ln \theta_{t-k} + \sum_{k=1}^{K} \gamma_{y,k}^h y_{t-k} + \gamma_x^h X_t + u_t^h$$

for monthly time t and horizon $h = 0, 1, \ldots, 36$ months. y_t is year-on-year CPI services inflation (excluding housing) in month t, computed as logarithmic changes. The dependent variable y_t^h denotes inflation h months ahead of time t. $\ln \theta_t$ is our independent variable of interest, the log vacancy-to-unemployment ratio from Barnichon (2010) and updated from JOLTS. X_t is a vector of control variables including labor productivity (from BLS), headline inflation shocks, and inflation expectations. Headline inflation shock is defined as the difference between year-on-year headline and core excluding food and energy inflation. Inflation expectations are measured using the 2-year inflation expectations from the Federal Reserve Bank of Cleveland, combined with 12-month inflation expectations from the Livingston survey before 1982. We include $K = 3$ lags of inflation and the log vacancy-to-unemployment ratio, with the auxiliary controls (labor productivity, the headline inflation shock, and inflation expectations) entering contemporaneously. The sample period is January 1986 to December 2023. Standard errors are Newey-West. The solid line shows the point estimates and the dashed lines show the 90 percent confidence interval. Source: BLS and authors' calculations.

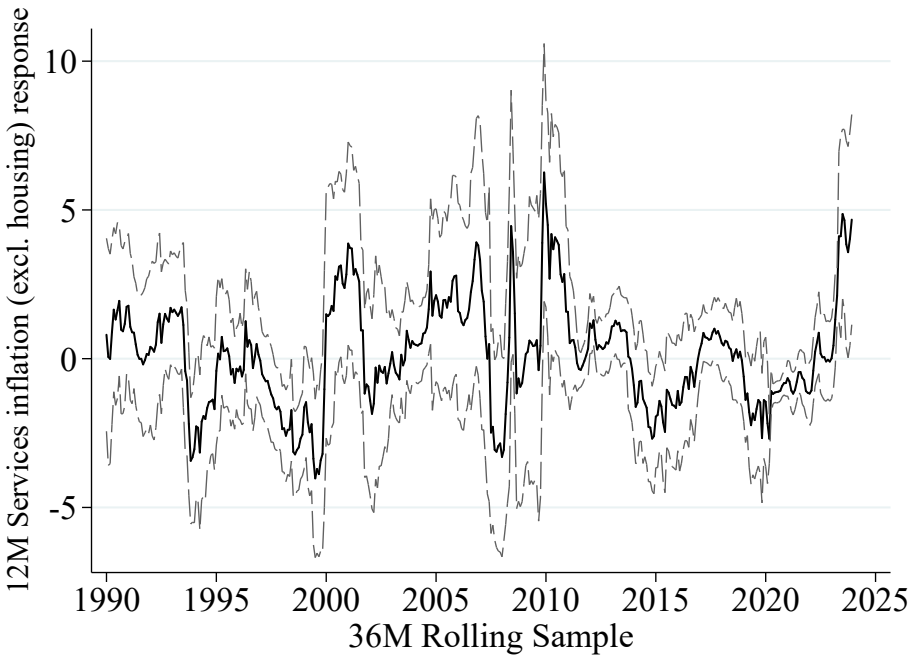


Figure 22: 12-month services inflation (excluding housing) response, 3-year rolling sample, national level official statistics

Notes: This figure plots rolling estimates from the following local projection (LP) model:

$$y_t^h = \alpha^h + \beta^h \ln \theta_t + \sum_{k=1}^{K} \gamma_{\theta,k}^h \ln \theta_{t-k} + \sum_{k=1}^{K} \gamma_{y,k}^h y_{t-k} + \gamma_x^h X_t + u_t^h$$

for monthly time t and horizon $h = 12$ months. The regression is estimated over rolling samples with a backward-looking 3-year window, and the x axis denotes the last month in each rolling sample. y_t is year-on-year CPI services inflation (excluding housing) in month t, computed as logarithmic changes. The dependent variable y_t^h denotes inflation h months ahead of time t. $\ln \theta_t$ is our independent variable of interest, the log vacancy-to-unemployment ratio from Barnichon (2010) and updated from JOLTS. X_t is a vector of control variables including labor productivity (from BLS), headline inflation shocks, and inflation expectations. Headline inflation shock is defined as the difference between year-on-year headline and core excluding food and energy inflation. Inflation expectations are measured using the 2-year inflation expectations from the Federal Reserve Bank of Cleveland, combined with 12-month inflation expectations from the Livingston survey before 1982. We include $K = 3$ lags of inflation and log V/U; labor productivity, the headline inflation shock, and inflation expectations enter contemporaneously. Standard errors are Newey-West. The solid line shows the point estimates and the dashed lines show the 90 percent confidence interval. Source: BLS and authors' calculations.

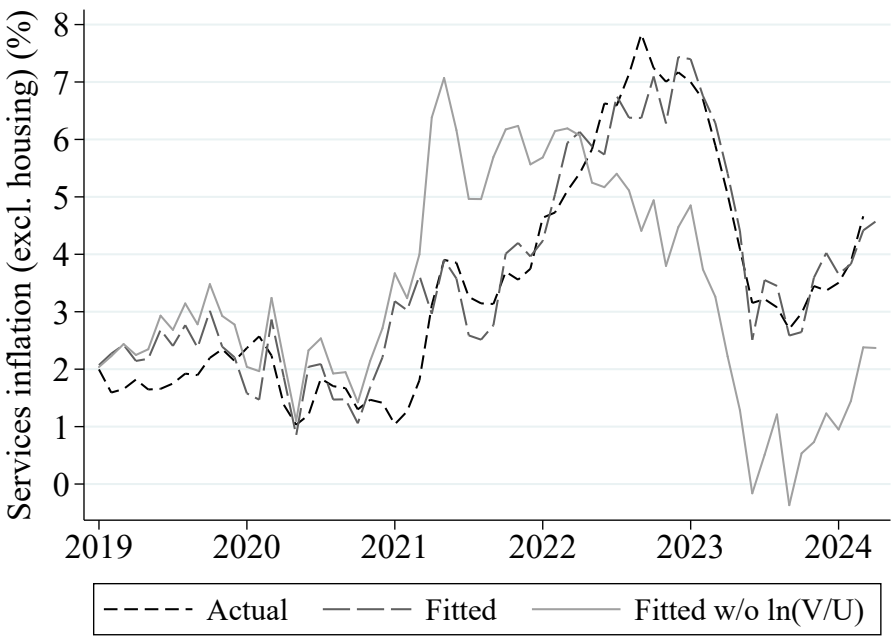


Figure 23: Predictions for services inflation (excluding housing), national level official statistics

Notes: This figure plots fitted values from the local projection (LP) model estimated over 3-year rolling samples (Figure 22):

$$y_t^h = \alpha^h + \beta^h \ln \theta_t + \sum_{k=1}^{K} \gamma_{\theta,k}^h \ln \theta_{t-k} + \sum_{k=1}^{K} \gamma_{y,k}^h y_{t-k} + \gamma_x^h X_t + u_t^h$$

for monthly time t and horizon $h = 12$ months. The regression is estimated over rolling samples with a backward-looking 3-year window. "Fitted" line plots predictions from the full model. "Fitted w/o ln(V/U)" line plots predictions from the same regression without a tight labor market by imposing that V/U equals 1. y_t is year-on-year CPI services inflation (excluding housing) in month t, computed as logarithmic changes. The dependent variable y_t^h denotes inflation h months ahead of time t. $\ln \theta_t$ is our independent variable of interest, the log vacancy-to-unemployment ratio from Barnichon (2010) and updated from JOLTS. X_t is a vector of control variables including labor productivity (from BLS), headline inflation shocks, and inflation expectations. Headline inflation shock is defined as the difference between year-on-year headline and core excluding food and energy inflation. Inflation expectations are measured using the 2-year inflation expectations from the Federal Reserve Bank of Cleveland, combined with 12-month inflation expectations from the Livingston survey before 1982. We include $K = 3$ lags of inflation and log V/U; labor productivity, the headline inflation shock, and inflation expectations enter contemporaneously. Source: BLS and authors' calculations.